

Contributions on robust distributed learning

Renaud Gaucher — Ph.D. Defense

Thursday, 3rd September 2026, Institut Henri Poincaré

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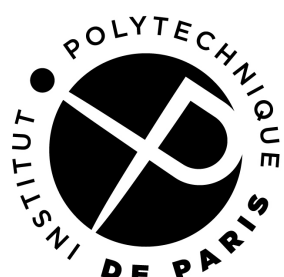
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ECOLE
DOCTORALE
DE MATHÉMATIQUES
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Outline

I. Introduction (Machine Learning, Optimization and Distributed Systems)

— Focus on three contributions —

II. Byzantine-robust distributed optimization

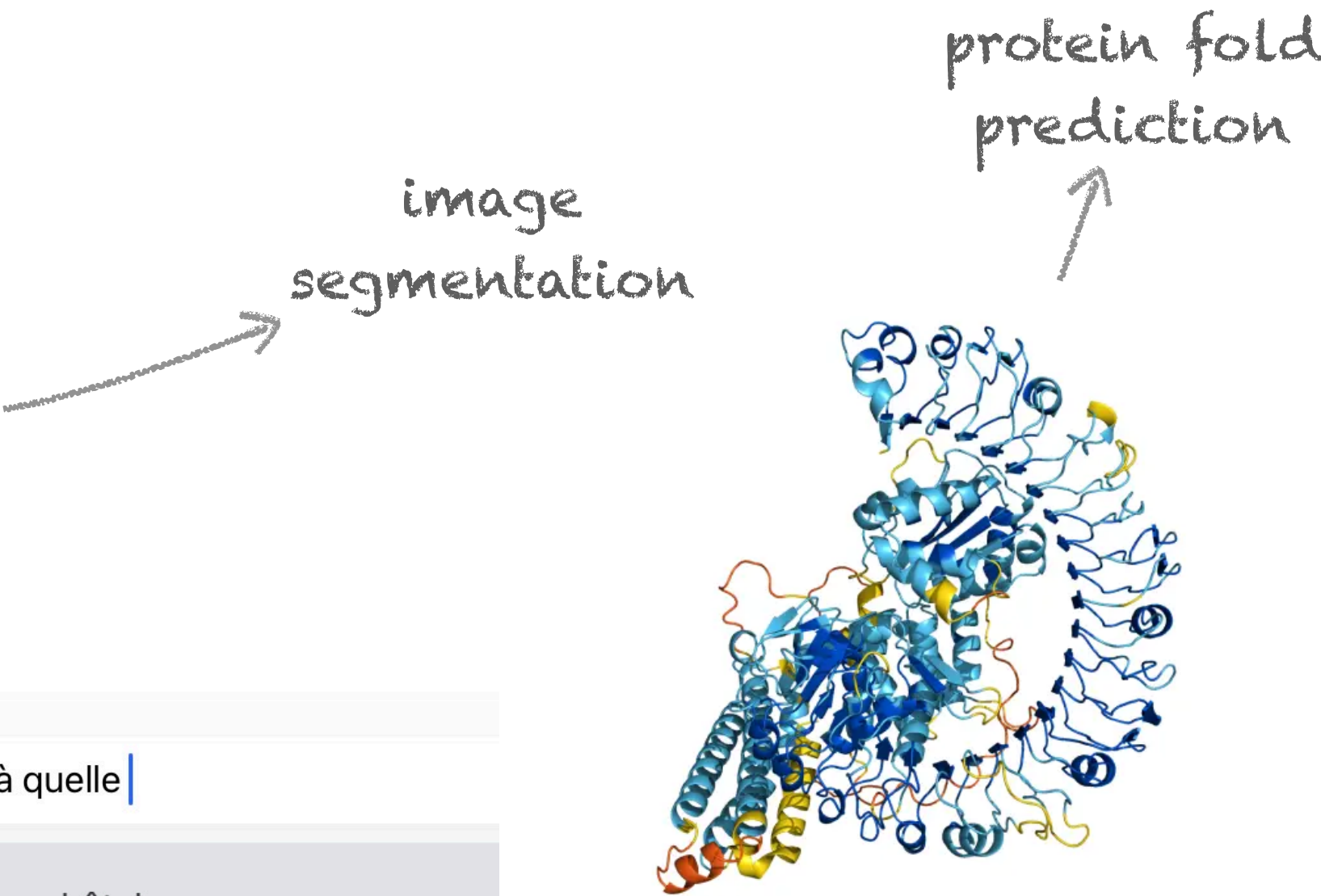
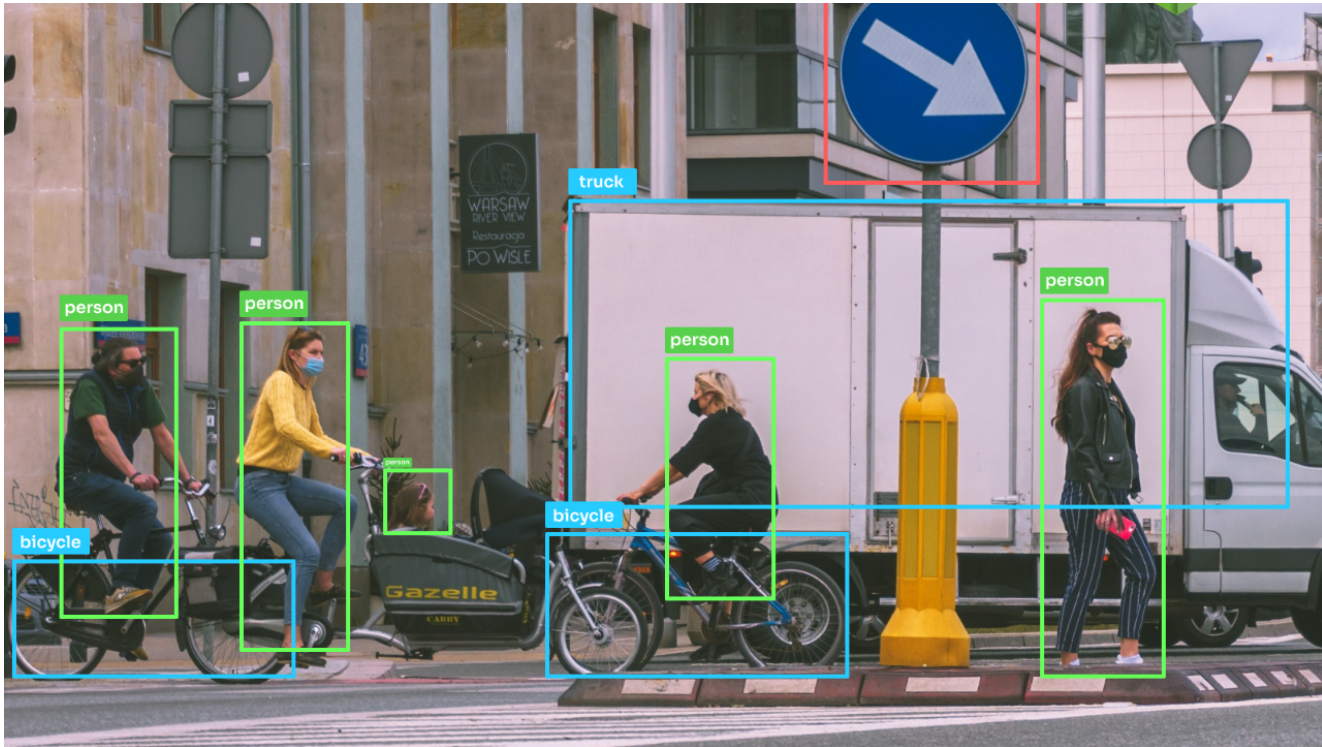
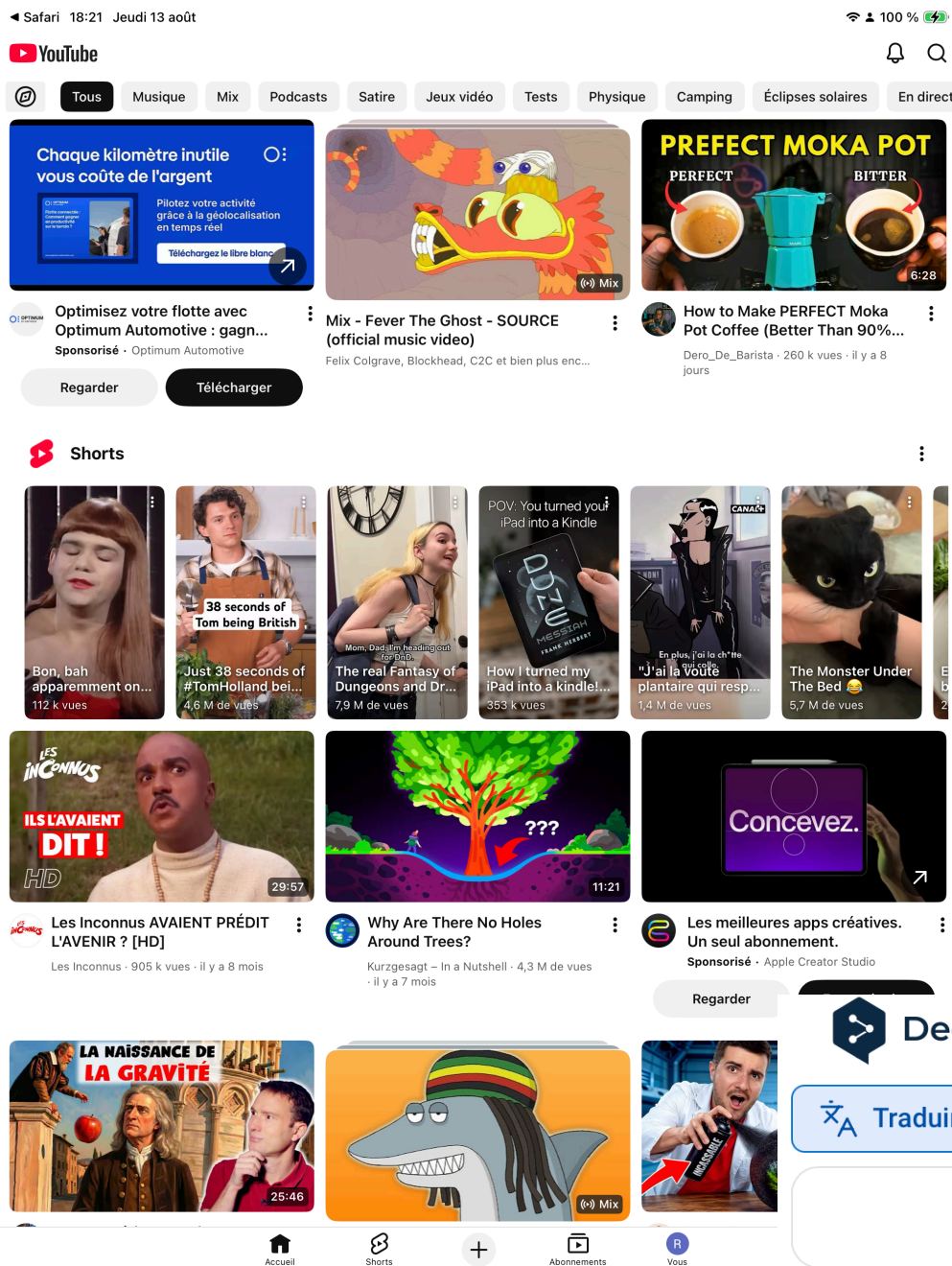
1. **The centralized setting** - *From Inexact Gradients to Byzantine Robustness: Acceleration and Optimization under Similarity*, **G.**, Dieuleveut and Hendrikx, 2026 (under review)
2. **The decentralized setting** - *A Unified Analysis of Byzantine Robust Gossip*, **G.**, Dieuleveut and Hendrikx, ICML 2025

III. Differential privacy

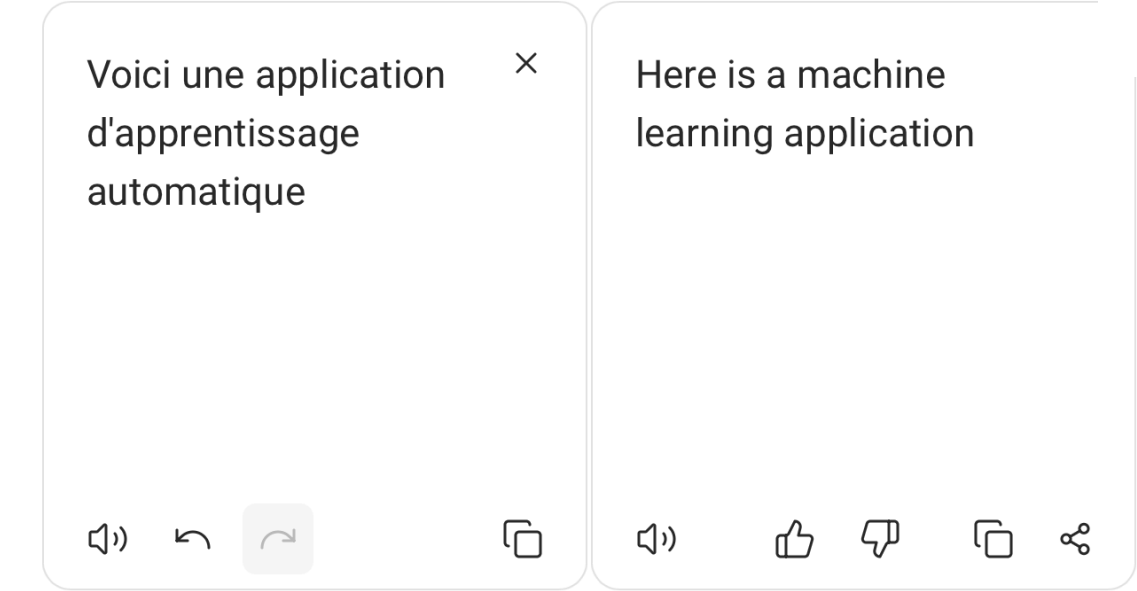
3. **Private mean estimation** - *The Curse of Dimensionality in Privacy: Anisotropic Private Mean estimation with Unknown Covariance*, **G.**, Linder, Smith, Ullman, Zakynthinou (to be submitted)

I - Introduction

Machine learning

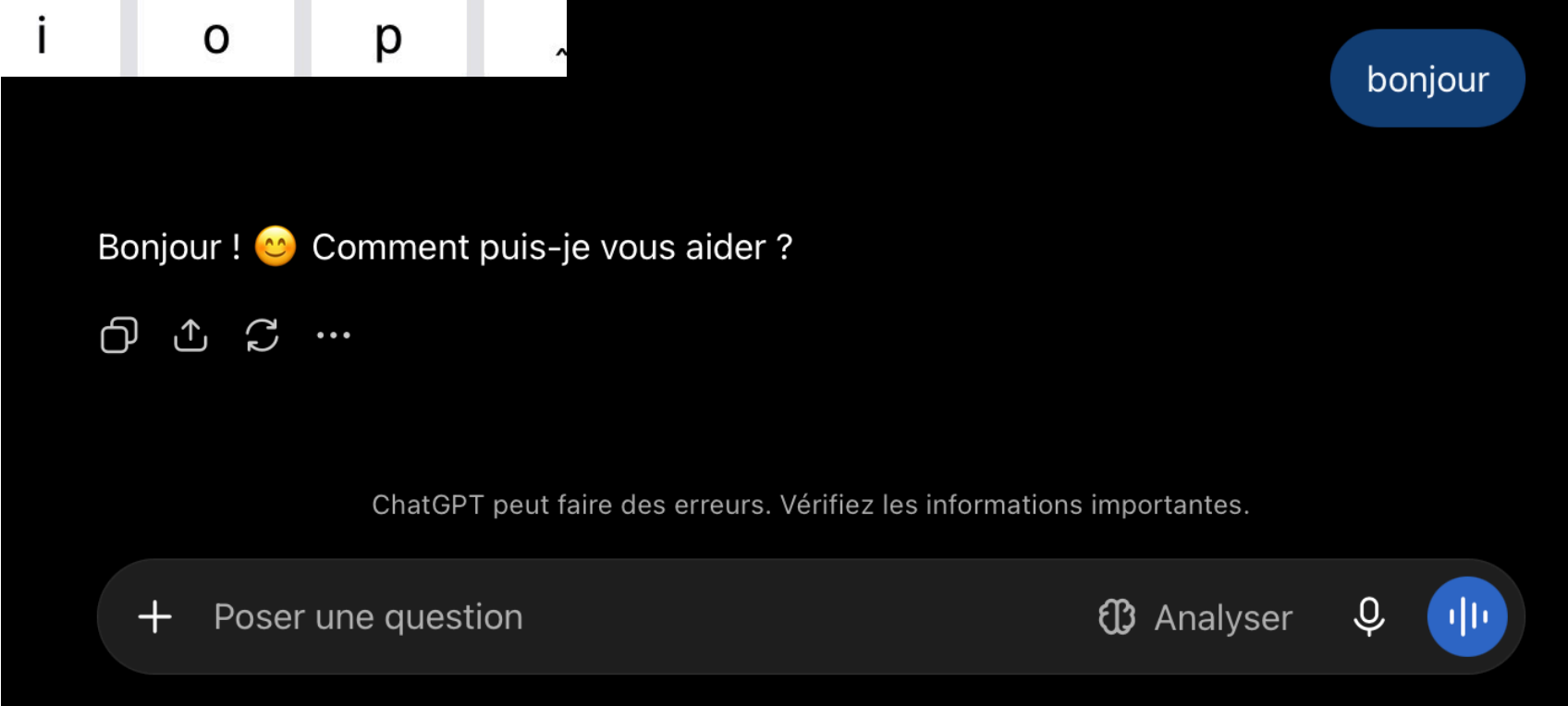


content and ad recommendation

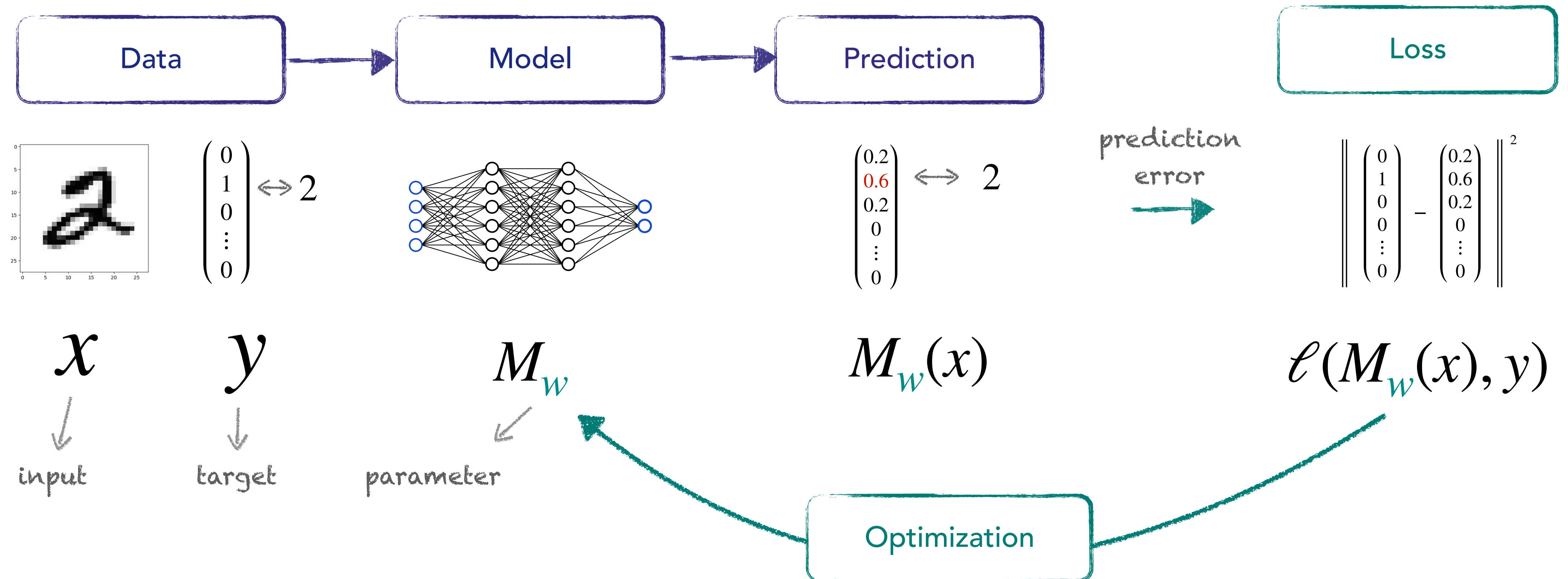


translation

next word prediction



Introduction to machine learning



Optimization: How to choose the model parameter w to minimize the loss ?

Introduction to optimization

Goal:

$$w^* = \arg \min_w \left\{ \mathcal{L}(w) = \sum_{(x,y) \in \mathcal{D}} \ell(M_w(x), y) \right\}$$

error on the whole dataset

Question: How fast is Gradient Descent ?

« Linear » convergence rate

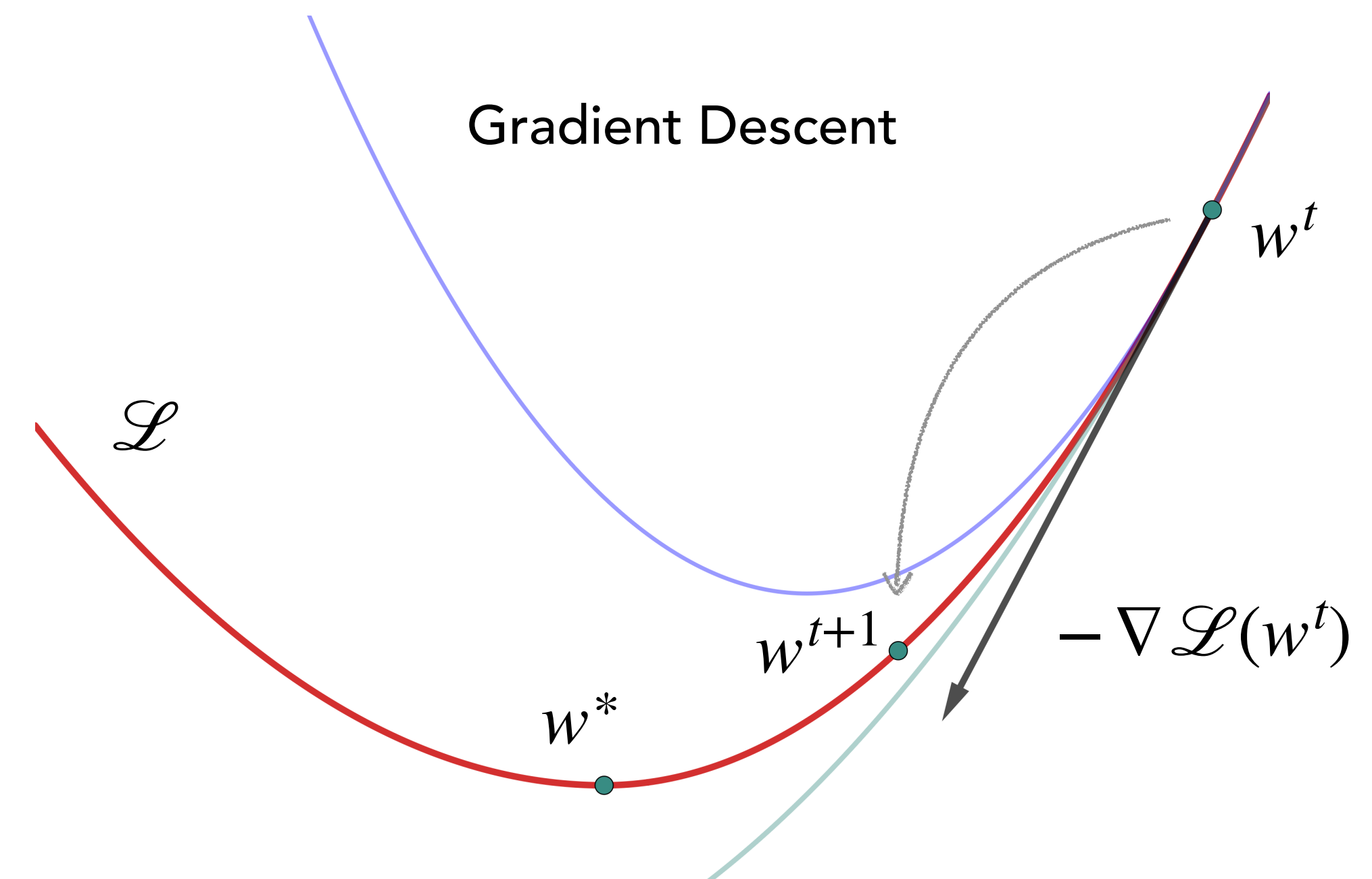
$$\|w^t - w^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|w^0 - w^*\|^2$$

iterations

hardness of the optimization
 L/μ : condition number

L - Smoothness assumption

$$\mathcal{L}(w) \leq \mathcal{L}(w^t) + \langle \nabla \mathcal{L}(w^t), w - w^t \rangle + \frac{L}{2} \|w - w^t\|^2$$



$$\mathcal{L}(w) \geq \mathcal{L}(w^t) + \langle \nabla \mathcal{L}(w^t), w - w^t \rangle + \frac{\mu}{2} \|w - w^t\|^2$$

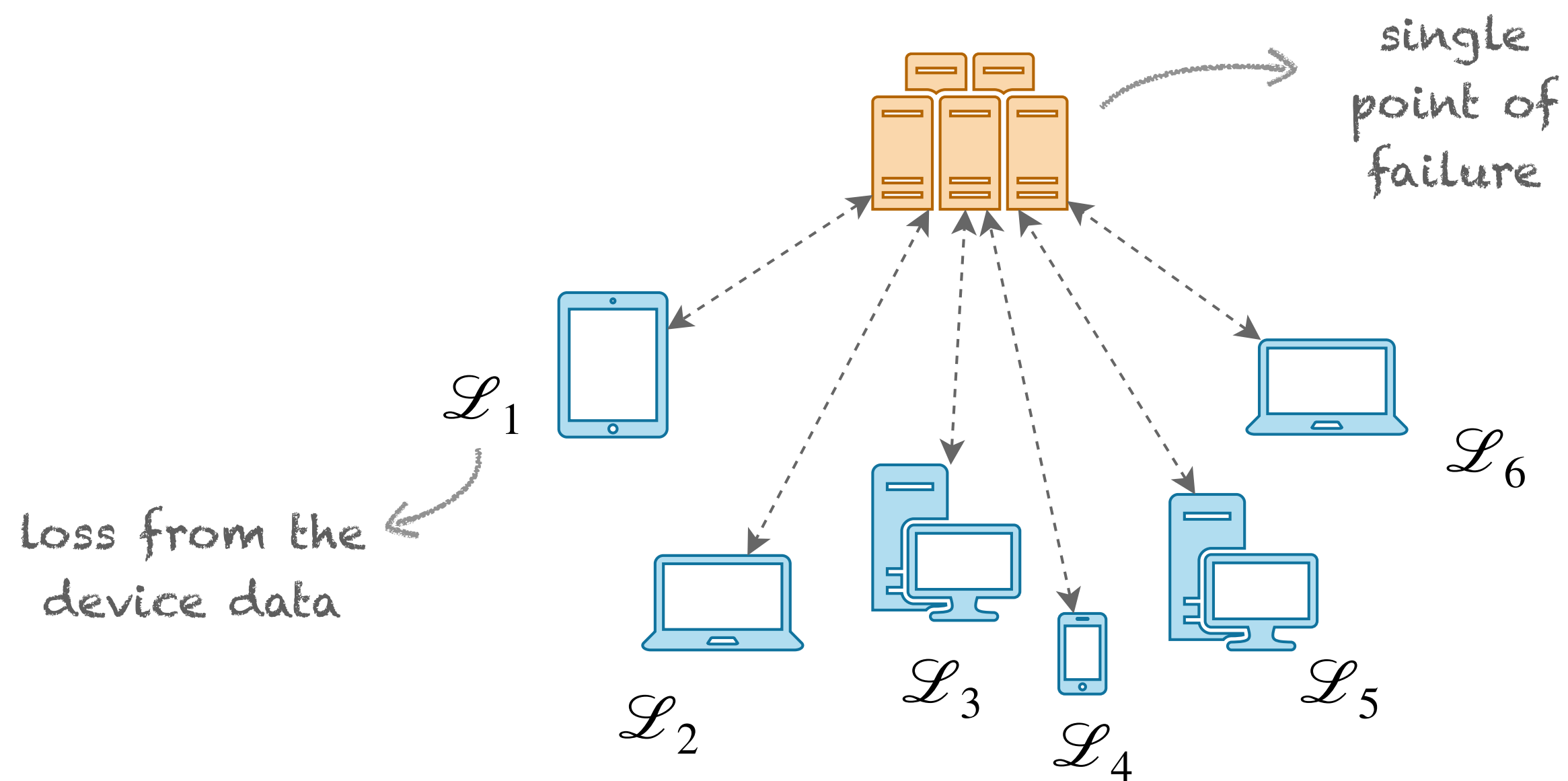
μ - Strong convexity assumption

Motivations and challenges of distributed optimization

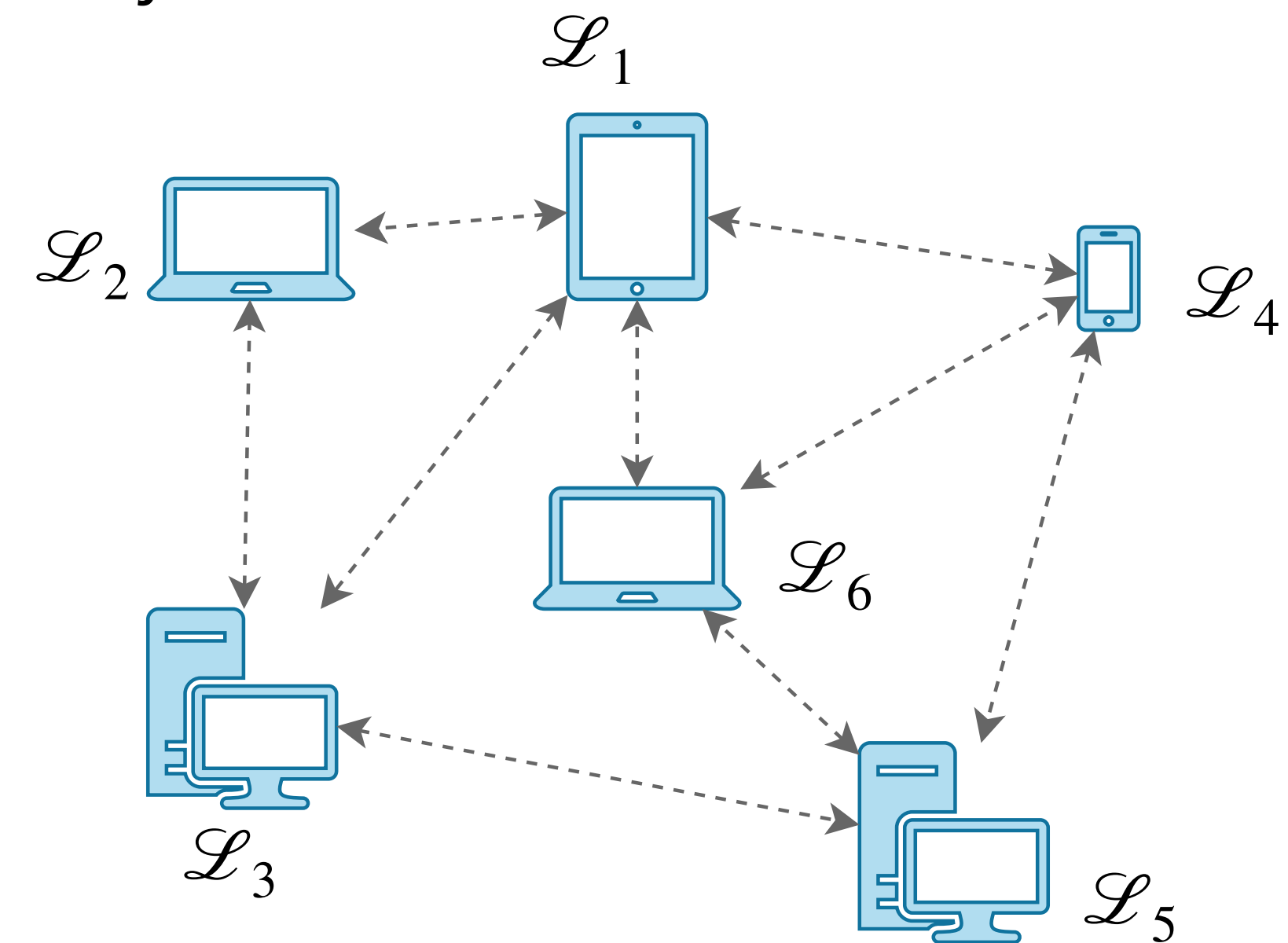
Motivations:

- Data sources or computing resources are distributed
- Keep the data stored on the user's device for privacy

$$\mathcal{L} = \frac{1}{n} \sum_{i \in [n]} \mathcal{L}_i$$



Centralized communication (server based)



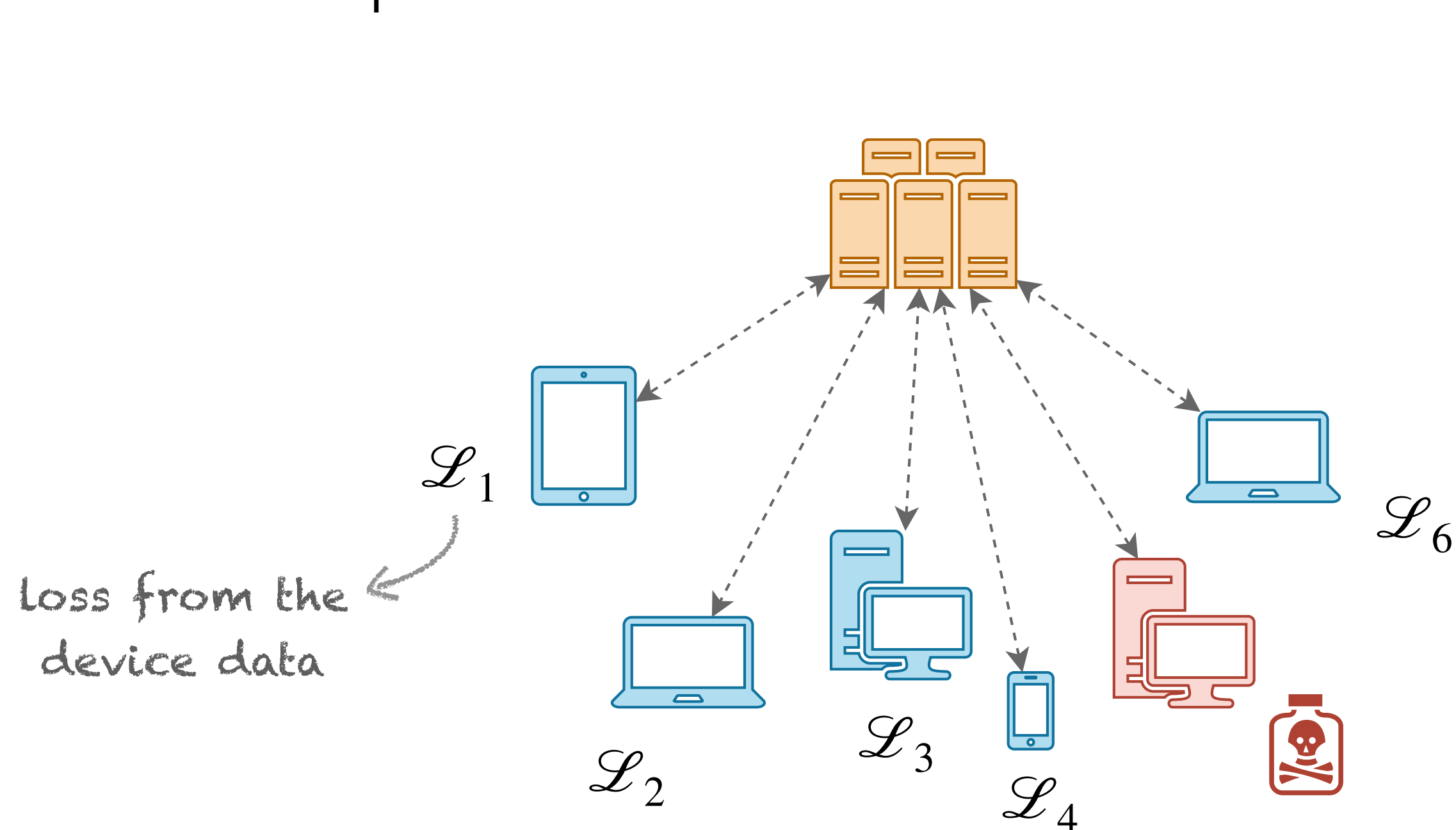
Decentralized communication (peer to peer)

Motivations and challenges of distributed optimization

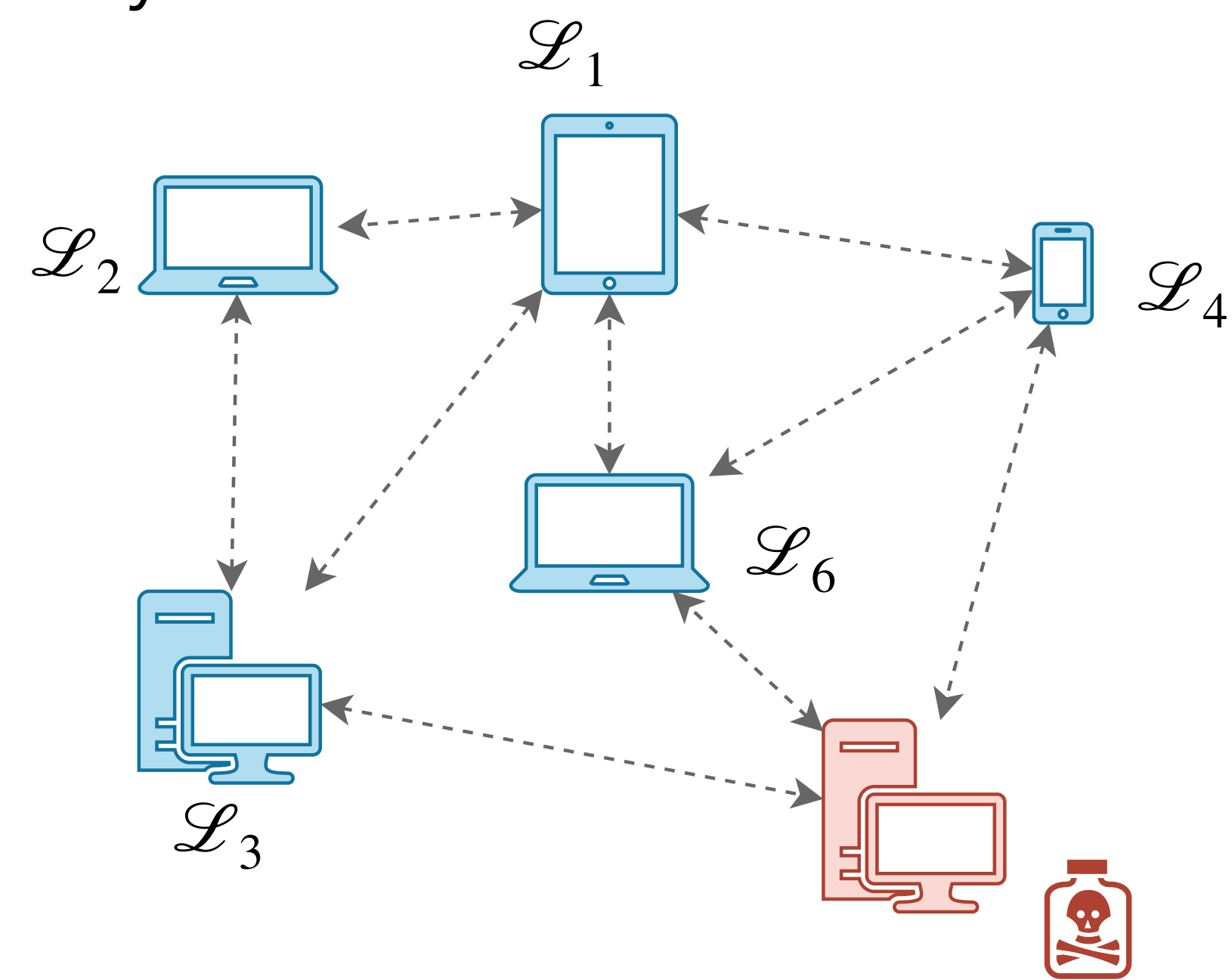
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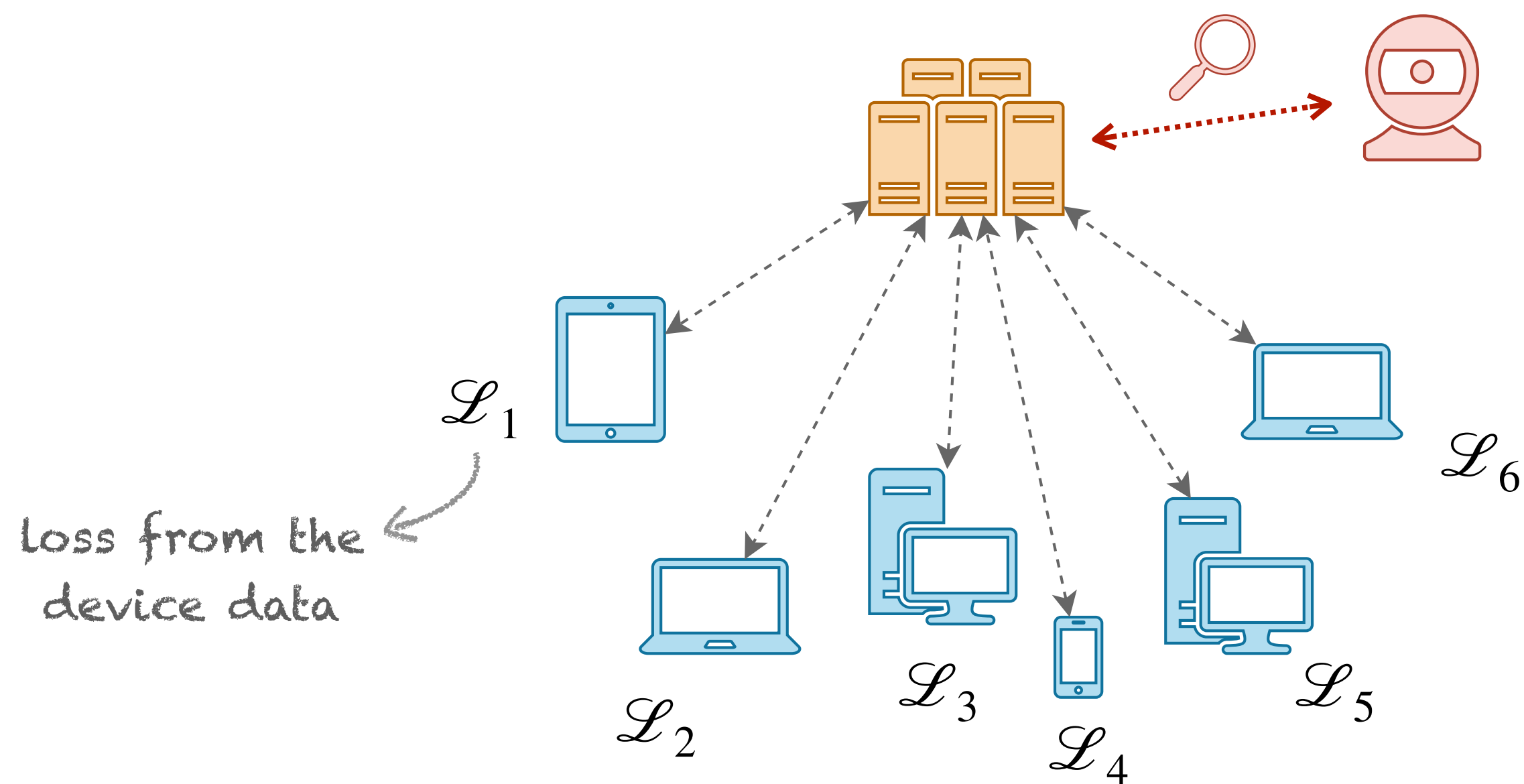
Threats: I) Corrupted devices (Byzantine failures)

Motivations and challenges of distributed optimization

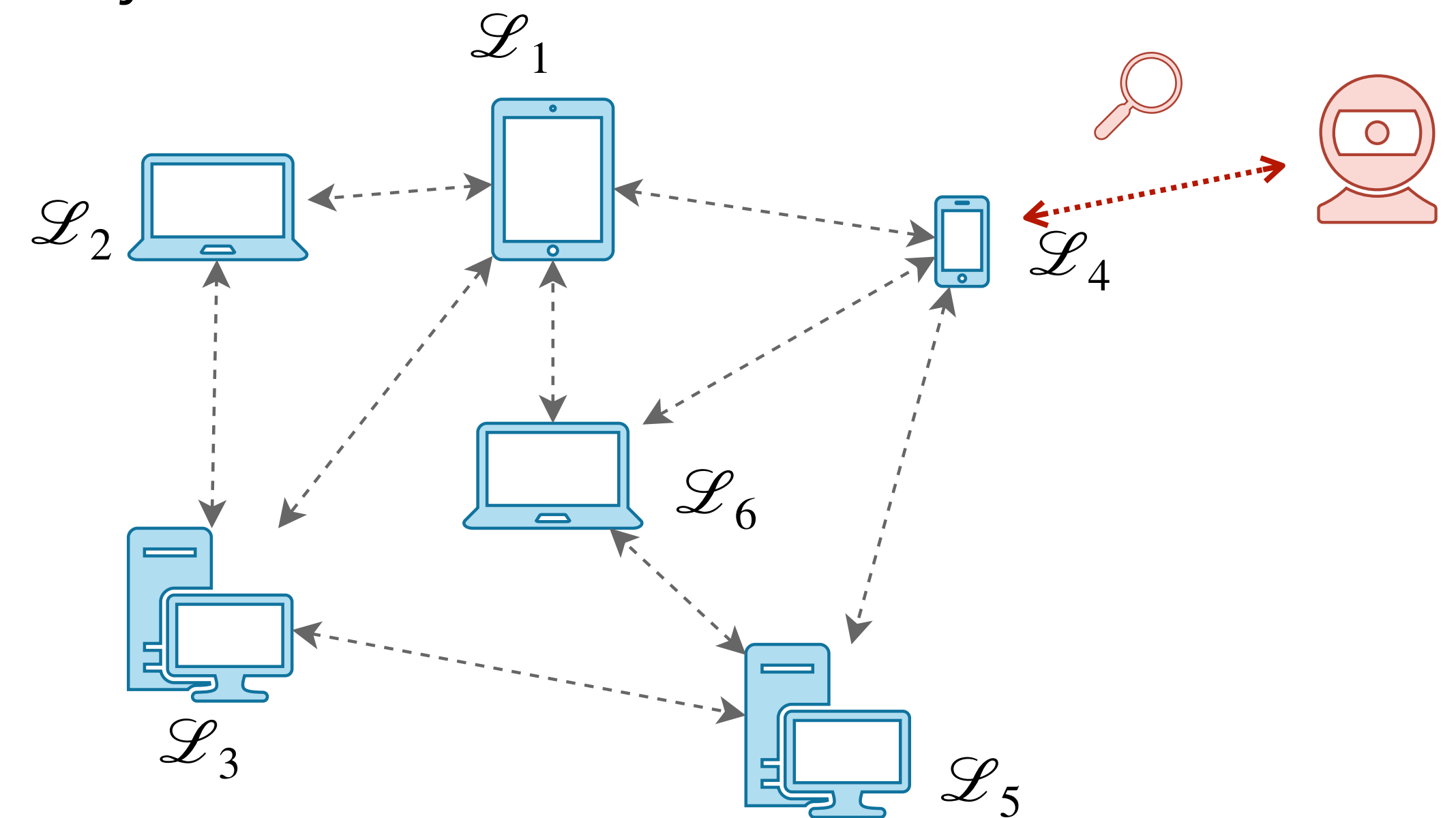
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$$\mathcal{L} = \frac{1}{n} \sum_{i \in [n]} \mathcal{L}_i$$



Centralized communication (server based)



Decentralized communication (peer to peer)

Threats: I) Corrupted devices (Byzantine failures) II) Sensitive Data disclosure (privacy)

I. Introduction

II. Byzantine-robust distributed optimization

1) The centralized setting

From Inexact Gradients to Byzantine Robustness: Acceleration and Optimization under Similarity,

G., Dieuleveut and Hendrikx, 2026 (under review)

- a. The inexact gradients formulation
- b. Acceleration
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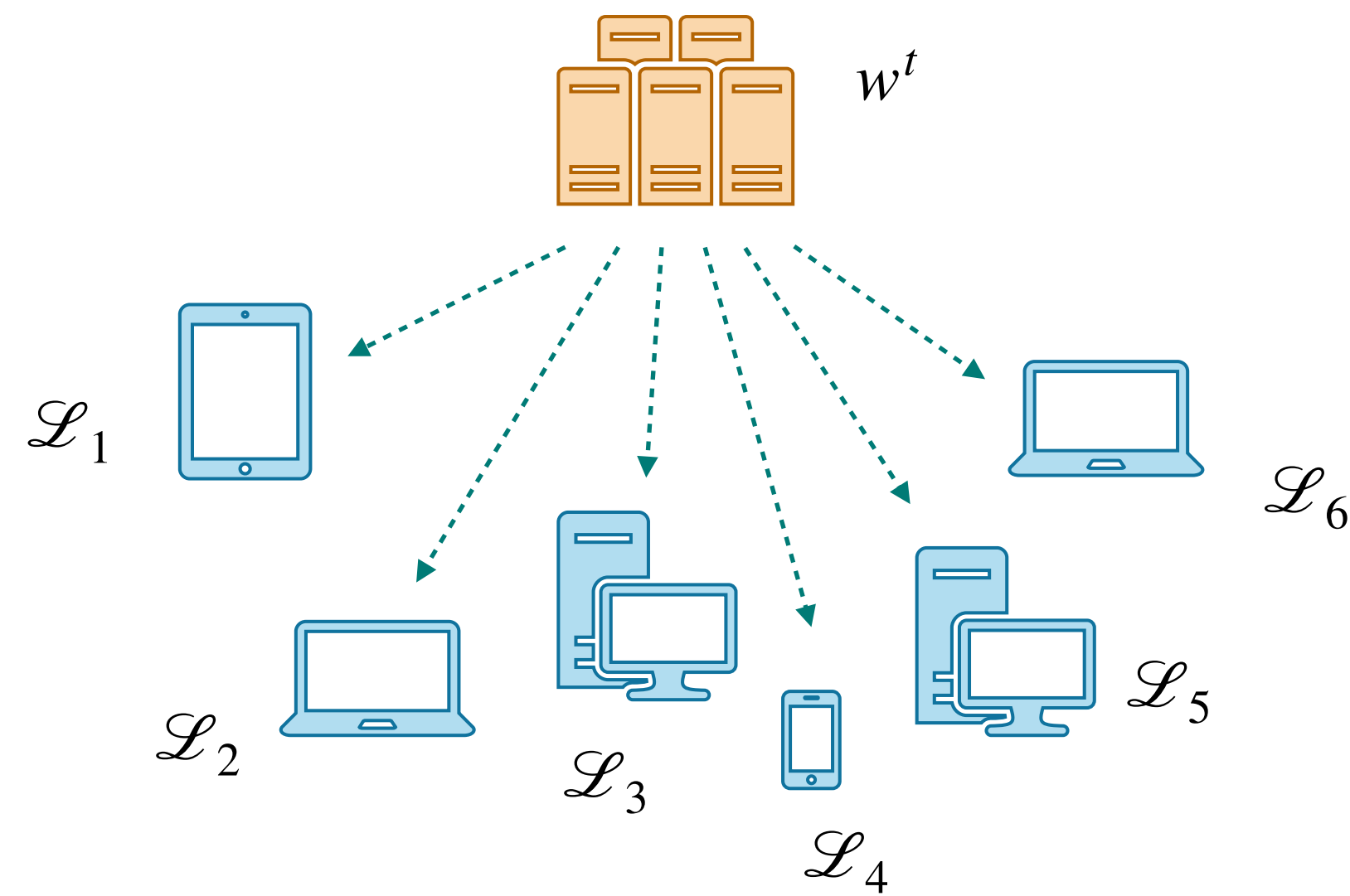
2) The decentralized setting

III. Differential privacy

IV. Conclusion and perspectives

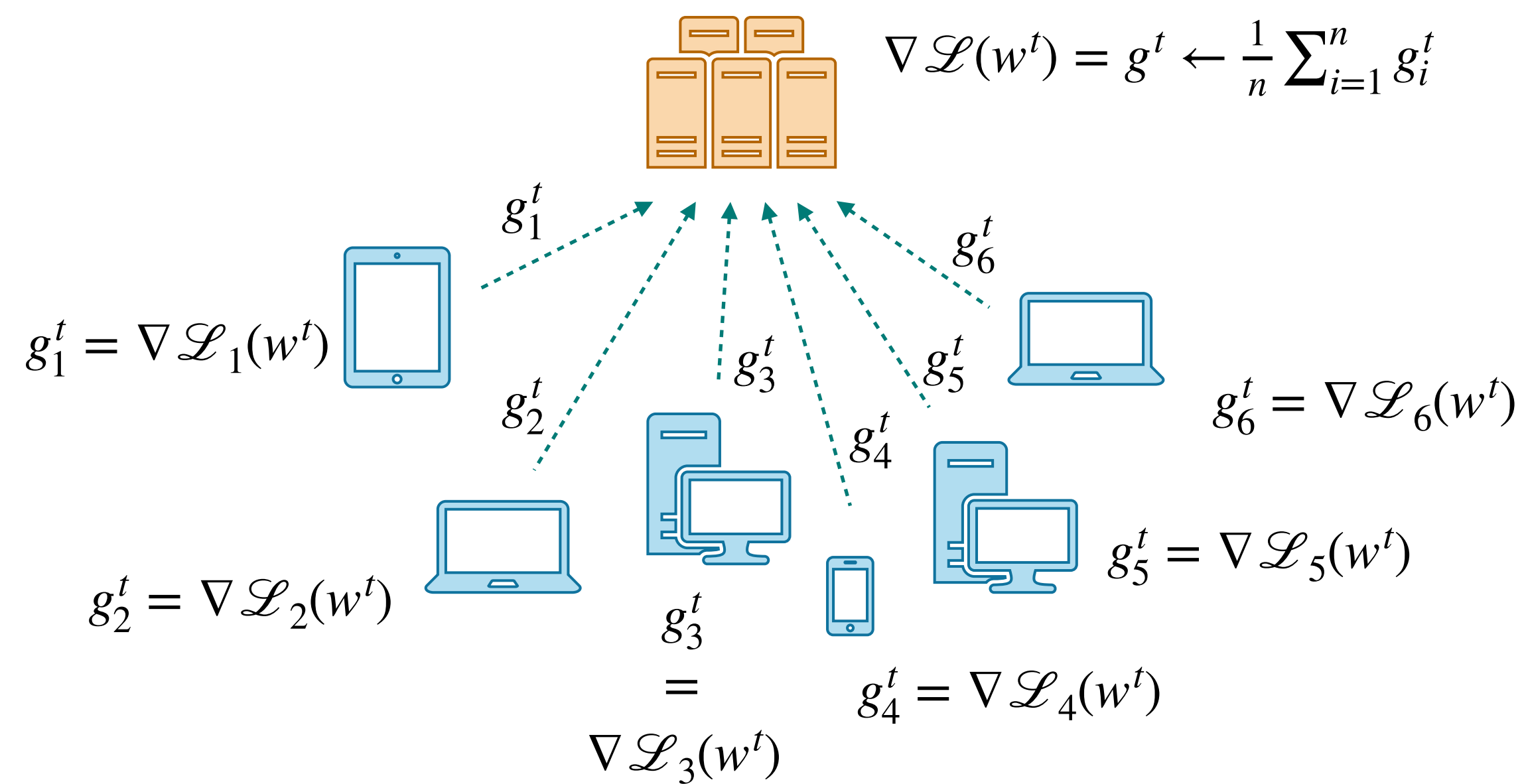
Basics of distributed optimization

$$\textbf{Goal: } \min_w \left\{ \mathcal{L}(w) = \frac{1}{n} \sum_{i \in [n]} \mathcal{L}_i(w) \right\}$$



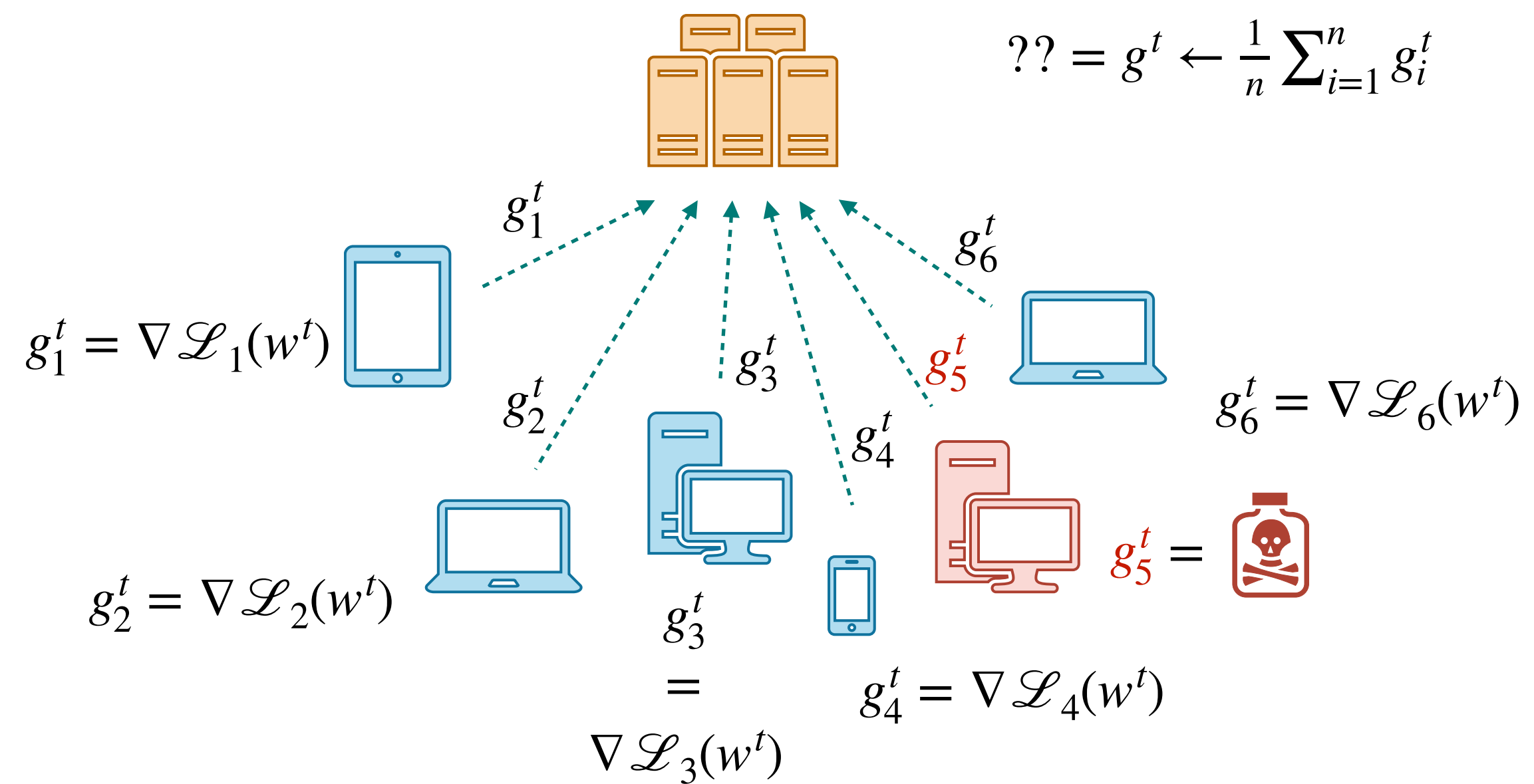
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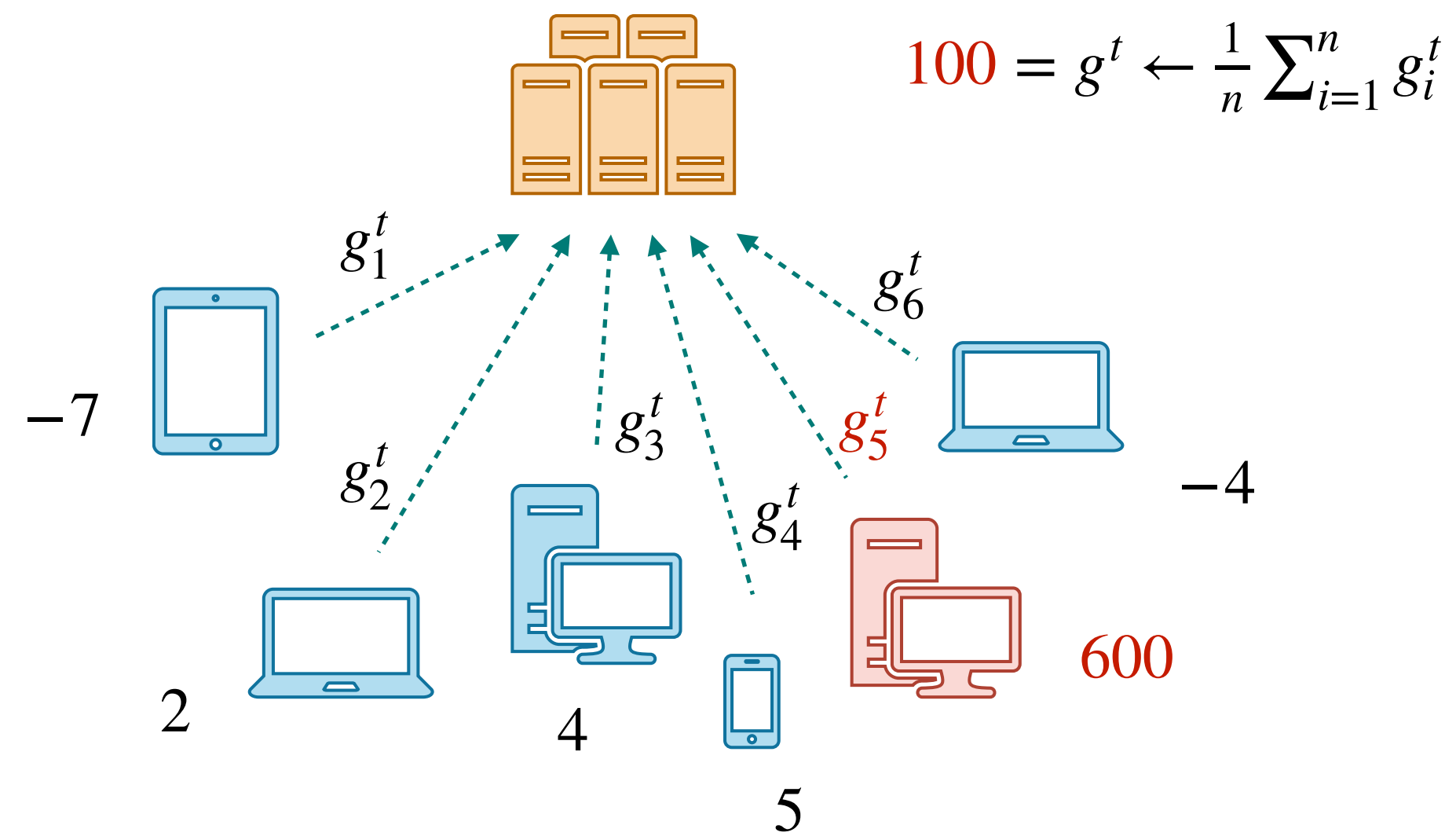
Basics of distributed optimization with f adversaries

$$\text{Goal: } \min_w \left\{ \mathcal{L}_h(w) = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \mathcal{L}_i(w) \right\}$$



Basics of distributed optimization with f adversaries

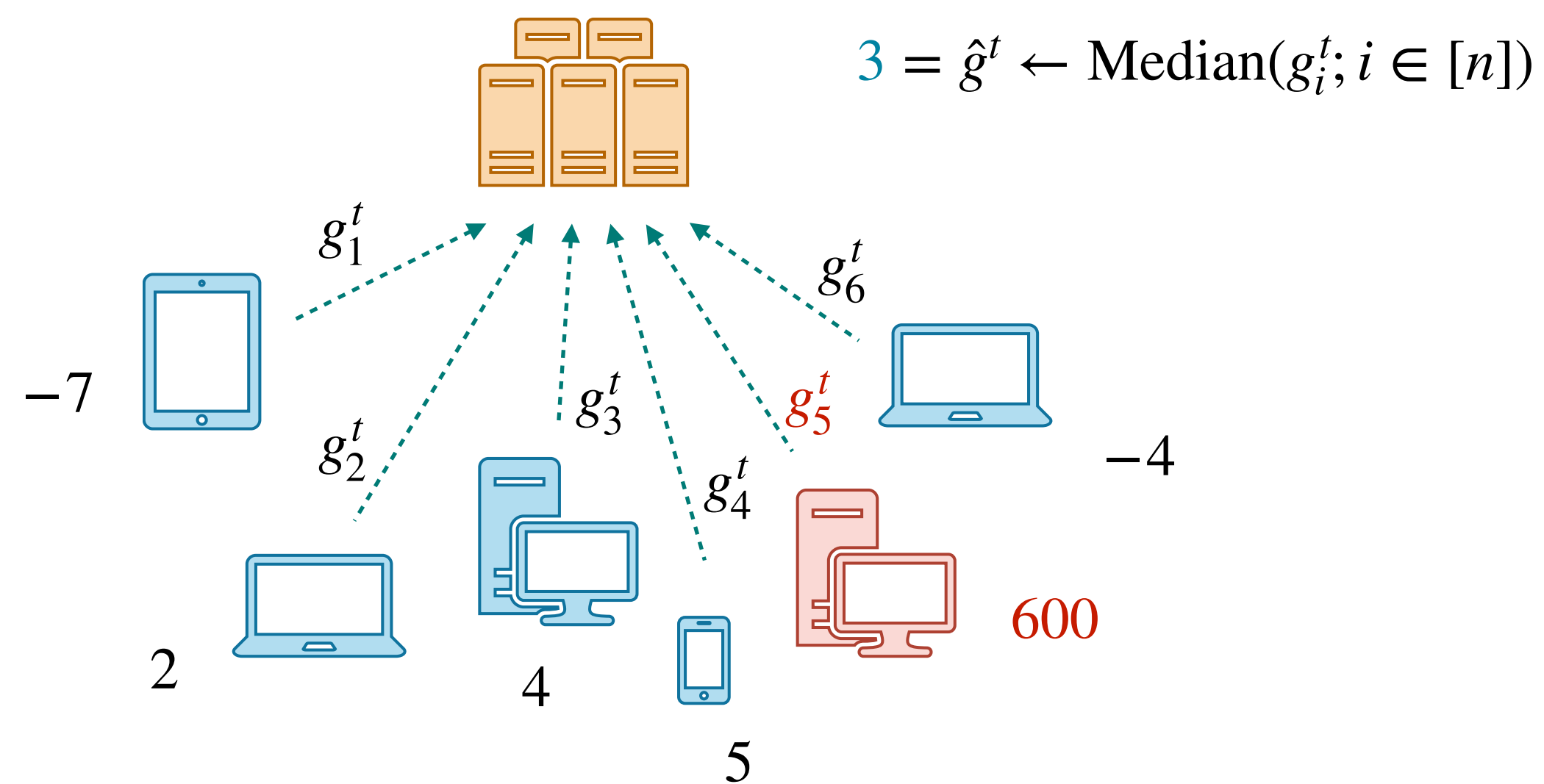
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⚠ Simply averaging the messages is not robust ⚠

Basics of distributed optimization with f adversaries

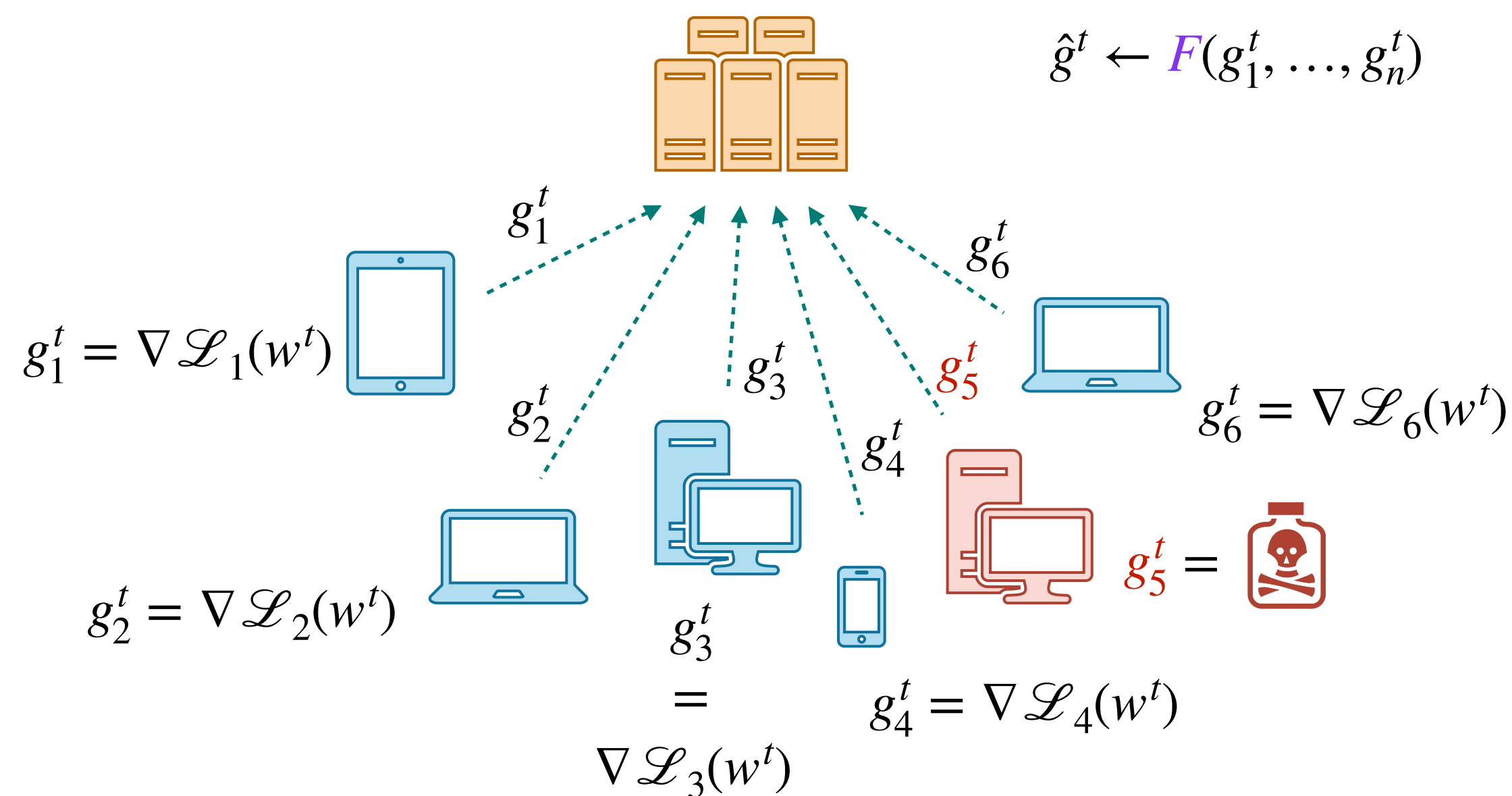
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⚠ Simply averaging the messages is not robust ⚠

Basics of distributed optimization with f adversaries

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! Simply averaging the messages is not robust !

Definition - (f, ν) robust aggregation (Allouha et al. 2023)

adversaries

For any $g_1, \dots, g_n \in \mathbb{R}^d$,

$$\|F(g_1, \dots, g_n) - \bar{g}_h\|^2 \leq \nu \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|g_i - \bar{g}_h\|^2$$

quality of F

$$\text{where } \bar{g}_h = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} g_i$$

- Geometric Median [Yin et al 2018]
- Coordinate-Wise Trimmed Mean [Yin et al 2018]
- Krum [Blanchard et al 2017]
- Huber-like estimator (G., Dieuleveut and Hendrikx, 2026)
- etc.

Aggregation	GM ^[1]	CWTM ^[1]	Krum ^[1]	Huber	Lower Bound ^[1]
ν	$4 \left(1 + \frac{f}{n-2f}\right)^2$	$6 \left(1 + \frac{6f}{n-2f}\right)$	$\frac{6f}{n-2f} \left(1 + \frac{6f}{n-2f}\right)$	$\frac{2f}{n-3f}$	$\frac{f}{n-2f}$

[1]: Allouha et al 2023

Robust average of gradients as an inexact gradient oracle

Assumption - (G,B) Heterogeneity

$(\mathcal{L}_i)_{i \in \text{honest}}$ satisfy

$$\frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|\nabla \mathcal{L}_i(w) - \nabla \mathcal{L}_h(w)\|^2 \leq G^2 + B^2 \|\nabla \mathcal{L}_h(w)\|^2$$

heterogeneity at the optimum

$\approx$ distance between Hessians $\nabla^2 \mathcal{L}_i - \nabla^2 \mathcal{L}_j$

Definition - (f, ν) robust aggregation (Allouha et al. 2023)

adversaries

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$$\|F(g_1, \dots, g_n) - \bar{g}_h\|^2 \leq \nu \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|g_i - \bar{g}_h\|^2$$

quality of F

where $\bar{g}_h = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} g_i$

Consequence (G., Dieuleveut & Hendrikx, 2026)

$\hat{g}(w) = F(\nabla \mathcal{L}_1(w), \dots, \nabla \mathcal{L}_n(w))$ is a (ζ^2, α) inexact gradient oracle:

$$\|\hat{g}(w) - \nabla \mathcal{L}_h(w)\|^2 \leq \zeta^2 + \alpha \|\nabla \mathcal{L}_h(w)\|^2, \text{ with } \zeta^2 = \nu G^2 \text{ and } \alpha = \nu B^2$$

the optimization literature
has studied inexact
gradients for >15 years

Question:

Can we achieve the optimal error with inexact gradients?

Tightness of the inexact gradient abstraction

▸ $\|\hat{g}(w) - \nabla \mathcal{L}_h(w)\|^2 \leq \zeta^2 + \alpha \|\nabla \mathcal{L}_h(w)\|^2$, with $\zeta^2 = \nu G^2$ and $\alpha = \nu B^2$

Question: When is $\hat{g}(w)$ non-informative?

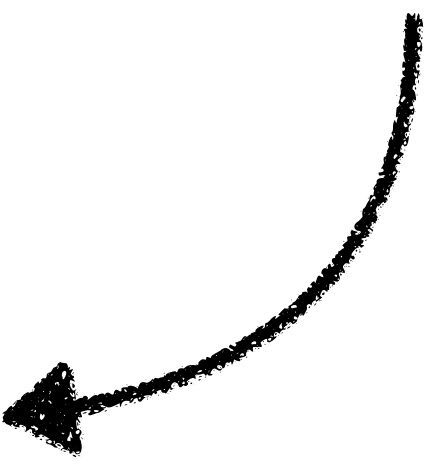
Answer: If $\hat{g}(w) = 0$ the optimization is stuck. It happens for $\|\nabla \mathcal{L}_h(w)\|^2 \leq \frac{\zeta^2}{1 - \alpha} = \frac{\nu G^2}{1 - \nu B^2}$

Theorem - Lower bound (Allouah et al., 2024)

Under (G, B) -Heterogeneity the best achievable error is

$$\|\nabla \mathcal{L}_h(\hat{w})\|^2 \approx \frac{f G^2}{n - (B^2 + 2)f} \text{ and } \mathcal{L}_h(\hat{w}) - \mathcal{L}_h(\hat{w}^*) \approx \frac{f G^2}{\mu(n - (B^2 + 2)f)}$$

Matches for ν optimal
i.e. $\nu = \frac{f}{n - 2f}$



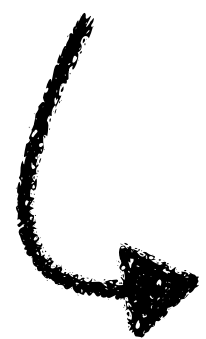
Theorem - Upper bound (Ajalloeian and Stitch, 2020)

Let $\mathcal{L}_h$ be μ -strongly convex and L -smooth, then GD with $(\zeta^2; \alpha)$ -inexact oracles and $\alpha < 1$ yields

$$\mathcal{L}_h(w^t) - \mathcal{L}_h^* \leq \left(1 - (1 - \alpha)\frac{\mu}{L}\right)^t (\mathcal{L}_h(w_0) - \mathcal{L}_h^*) + \frac{\zeta^2}{\mu(1 - \alpha)} \rightarrow \text{optimal error}$$

Tightness of the inexact gradient abstraction

The inexact gradients formulation is tight for Byzantine-robust optimization



We focus on optimization methods based on inexact gradients

Since one computation of $\hat{g}(w)$ = one communication:

Question:

Can we communicate less than with distributed gradient descent ?

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- a. The inexact gradients formulation
- b. Acceleration**
- c. Optimization under Similarity

2) The decentralized setting

III. Differential privacy

IV. Conclusion and perspectives

Fast algorithms using inexact gradients

Question:

Can we have the optimal asymptotic error and acceleration?

$$\mathcal{L}_h(w^t) - \mathcal{L}_h^* \leq \text{Optimization Error} + \text{Asymptotic Error}$$

	(ζ^2, α) -Inexact gradient method	Oracles call required for Optimization Error $\leq \varepsilon$	Asymptotic Error	Condition on α
	GD	$\Theta\left(\frac{L}{\mu} \log(1/\varepsilon)\right)$	$\frac{\zeta^2}{\mu(1-\alpha)}$	$\alpha < 1$
non-accelerated rate ←	SOTA accelerated methods [2]	$\Theta\left(\sqrt{\frac{L}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\sqrt{\frac{L}{\mu}} \frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \frac{\mu}{L}$
accelerated rate ←	G., Dieuleveut & Hendriks, 2026	$\Theta\left(\sqrt{\frac{L}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \sqrt{\frac{\mu}{L}}$

[2] cf. Schmidt et al. 2011, Devolder et al 2014, Vasin et al. 2023, Kornilov et al. 2025, Vasin et al. 2025

An accelerated Inexact Extra-gradient Method

Theorem (G., Dieuleveut & Hendriks 2026)

Let $\mathcal{L}_h$ be μ -strongly convex and L -smooth, and $\hat{g}$ a (ζ^2, α) -inexact gradient oracles with $\alpha < 1/16$. For $\theta = 1/2^{3/2}L$, $\eta = \min\left(\frac{1}{2\mu}, \frac{\theta}{24\tau}\right)$, and

$$\tau = \min\left(\frac{\theta\mu}{16\alpha}, \sqrt{\frac{\theta\mu}{48}}\right), \text{ we have}$$

$$\mathcal{L}_h(w_f^T) - \mathcal{L}_h(w^*) \leq O\left((1 - \rho)^T R + \frac{\zeta^2}{\mu}\right)$$

where

$$\rho = \Theta\left(\min\left(\sqrt{\frac{\mu}{L}}, \frac{\mu}{\alpha L}\right)\right) \text{ and } R = \tau^2 \Delta \|w^0 - w^*\|^2 + \mathcal{L}_h(w^0) - \mathcal{L}_h(w^*)$$

accelerated rate
if $\alpha \lesssim \sqrt{\mu/L}$

Accelerated Inexact Extra-gradient^[3]

Input: $w^0 = w_f^0$, oracle sampler $\tilde{\nabla} \mathcal{L}$

Parameter: $\tau \in (0,1)$, η , θ , K

For $t = 0, \dots, T-1$ do

$$\triangleright w_g^t = \tau w^t + (1 - \tau) w_f^t$$

$$\triangleright w_f^{t+1} = w_g^t - \theta \hat{g}(w_g^t)$$

$$\triangleright w^{t+1} = w^t + \eta \mu (w_f^{t+1} - w^t) - \eta \hat{g}(w_f^{t+1})$$

End For

Output: w_f^T

optimal error

Fast algorithms using inexact gradients

Question:

Can we go faster than GD

when $\alpha \gg \sqrt{\frac{\mu}{L}}$?

$$\mathcal{L}_h(w^t) - \mathcal{L}_h^* \leq \text{Optimization Error} + \text{Asymptotic Error}$$

	(ζ^2, α) -Inexact gradient method	Oracles call required for Optimization Error $\leq \varepsilon$	Asymptotic Error	Condition on α
non-accelerated rate	GD	$\Theta\left(\frac{L}{\mu} \log(1/\varepsilon)\right)$	$\frac{\zeta^2}{\mu(1-\alpha)}$	$\alpha < 1$
	SOTA accelerated methods [2]	$\Theta\left(\sqrt{\frac{L}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\sqrt{\frac{L}{\mu}} \frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \frac{\mu}{L}$
accelerated rate	G. , Dieuleveut & Hendriks, 2026 (acc. extra grad)	$\Theta\left(\sqrt{\frac{L}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \sqrt{\frac{\mu}{L}}$

[2] cf. Schmidt et al. 2011, Devolder et al 2014, Vasin et al. 2023, Kornilov et al. 2025, Vasin et al. 2025

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Leveraging a proxy of the loss function

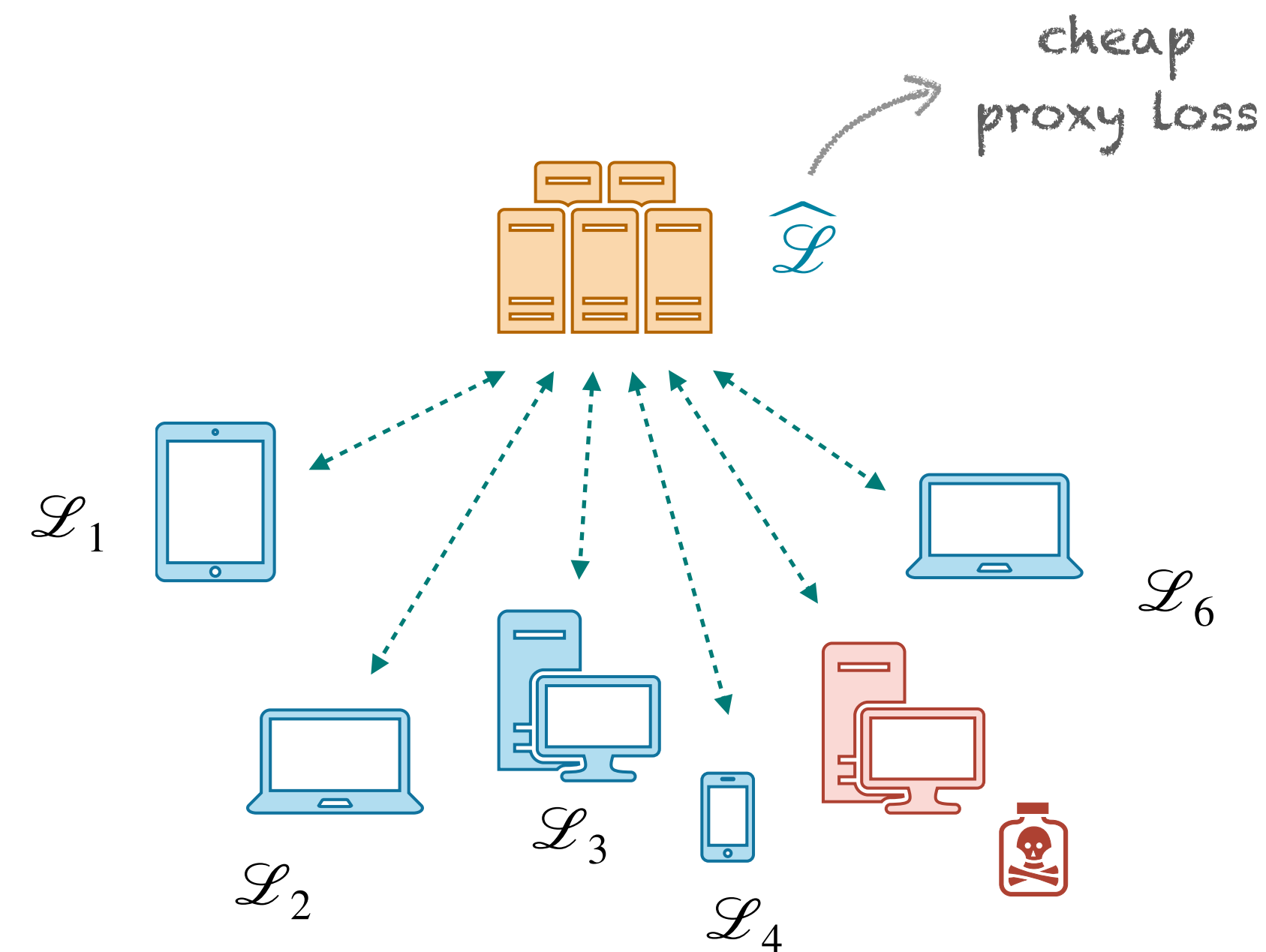
Idea: change the topology to reduce the condition number L/μ

Forward-backward splitting $\mathcal{L}_h = (\mathcal{L}_h - \widehat{\mathcal{L}}) + \widehat{\mathcal{L}}$

forward step $\leftarrow$ backward step $\rightarrow$

Proximal Inexact Gradient under Similarity (PIGS)

$$w_{t+1} = \arg \min_w \underbrace{\langle \hat{g}(w_t) - \nabla \widehat{\mathcal{L}}(w_t), w \rangle + \frac{1}{2\eta} \|w - w_t\|^2}_{\text{small smoothness constant}} + \underbrace{\widehat{\mathcal{L}}(w)}_{\text{cheap proxy}}$$



An alternative notion of inexact gradients is needed

Definition - $(\zeta^2, \tilde{\alpha})$ -Inexact oracles

For $\mathcal{L}_h$ μ -strongly convex

$$\|\hat{g}(w) - \nabla \mathcal{L}_h(w)\|^2 \leq \zeta^2 + \tilde{\alpha}\mu^2\|w - w^*\|^2$$



$$\text{implies } \|\hat{g}(w) - \nabla \mathcal{L}_h(w)\|^2 \leq \zeta^2 + \tilde{\alpha}\|\nabla \mathcal{L}_h(w)\|^2$$

Assumption - Pairwise Hessian similarity

For any i, j **honest**

$$\|\nabla^2 \mathcal{L}_i(w) - \nabla^2 \mathcal{L}_j(w)\|_{op} \leq \Delta$$

Proposition (G., Dieuleveut & Hendrikx 2026)

If also $(\mathcal{L}_i)_{i \in \text{honest}}$ are convex, $\hat{g}(w) = F(\nabla \mathcal{L}_1(w), \dots, \nabla \mathcal{L}_1(w))$ is a $(\zeta^2, \tilde{\alpha})$ -inexact oracle with

$$\zeta^2 = \nu \frac{2}{|\text{honest}|} \sum_{i \in \text{honest}} \|\nabla \mathcal{L}_i(w^*)\|^2 \quad \text{and} \quad \tilde{\alpha} = \nu \frac{2\Delta^2}{\mu^2}$$

Convergence of Proximal Inexact Gradient with Similarity

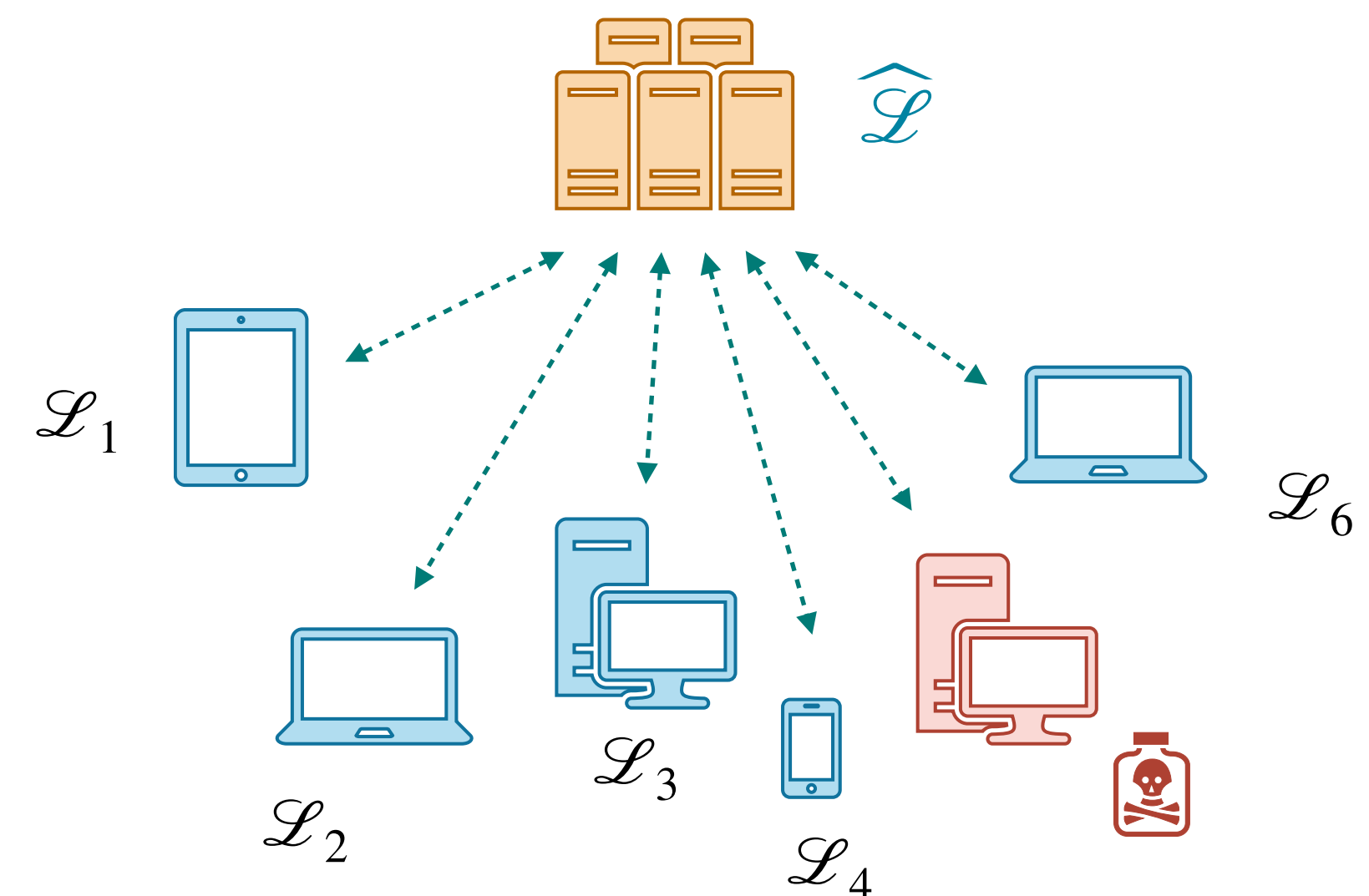
Theorem - Convergence of PIGS (G., Dieuleveut & Hendrikx 2026)

Assume $\hat{g}(w)$ is a $(\zeta^2, \tilde{\alpha})$ inexact oracle with $\tilde{\alpha} < 1/4$, and $\mathcal{L}_h - \widehat{\mathcal{L}}$ is Δ -smooth

$$\mathcal{L}_h(w_T) - \mathcal{L}_h(w^*) \lesssim \left(1 - \frac{\mu}{12\Delta}\right)^T \frac{\Delta \|w_0 - w^*\|^2}{(1 - 4\tilde{\alpha})} + \frac{\zeta^2}{\mu(1 - 4\tilde{\alpha})}$$

independent of L

Question: to what extent is Δ/μ smaller than L/μ ?



Convergence of Proximal Inexact Gradient with Similarity

Theorem - Convergence of PIGS (G., Dieuleveut & Hendrikx 2026)

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optimal error

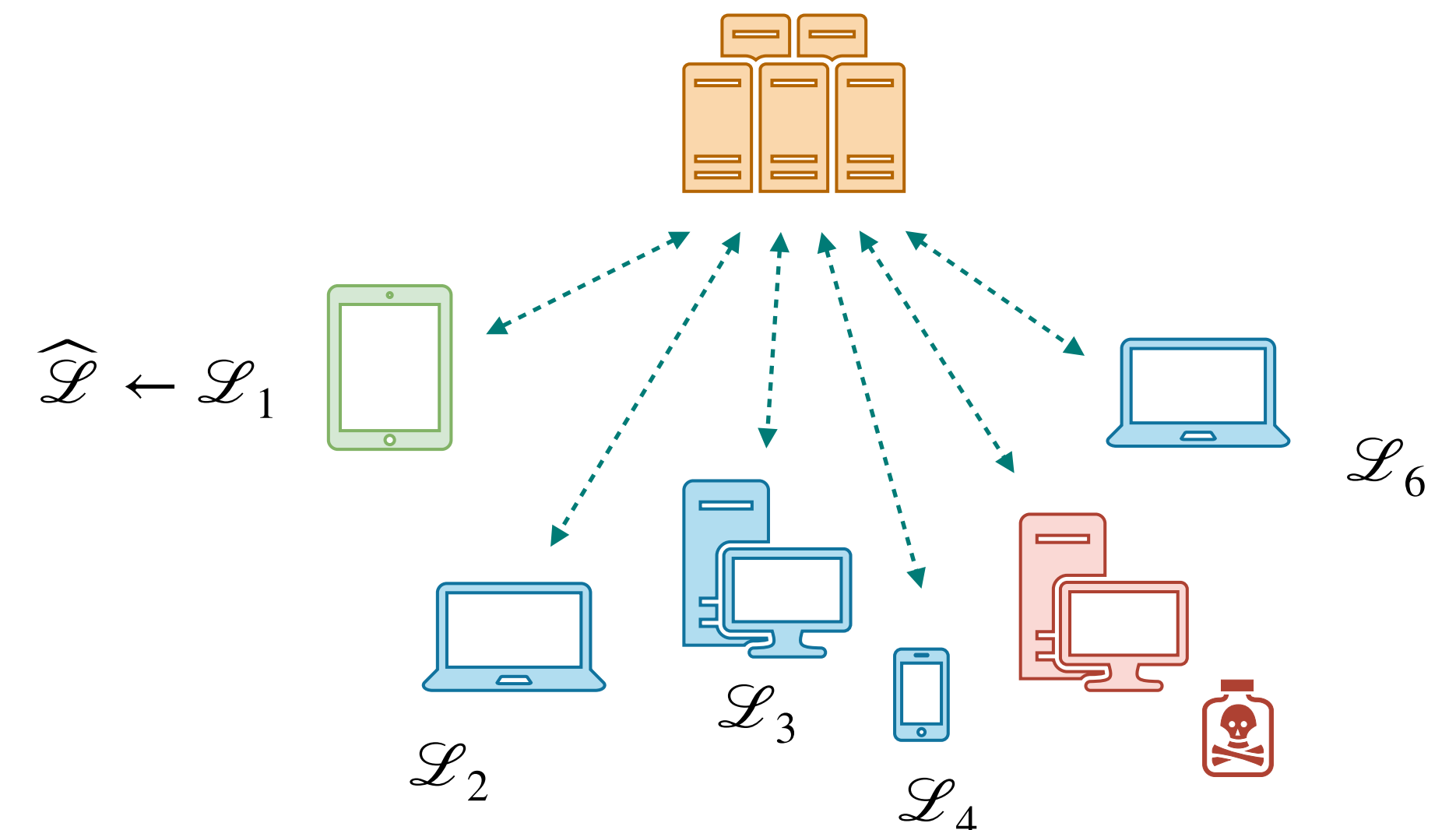
independent of L

Question: to what extent is Δ/μ smaller than L/μ ?

→ Let $\widehat{\mathcal{L}} \leftarrow \mathcal{L}_1$ in empirical risk minimisation:^[4]

$$\frac{\Delta}{\mu} = \tilde{O}\left(\frac{L}{\mu} \frac{1}{\sqrt{|\mathcal{D}_i|}}\right)$$

local dataset



Combining acceleration and similarity

Question:

Can we have acceleration and similarity ?

$\mathcal{L}_h(w^t) - \mathcal{L}_h^* \leq \text{Optimization Error} + \text{Asymptotic Error}$			
(ζ^2, α) -Inexact gradient method	Oracles call required for Optimization Error $\leq \varepsilon$	Asymptotic Error	Condition on α
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????	$\Theta\left(\sqrt{\frac{\Delta}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	

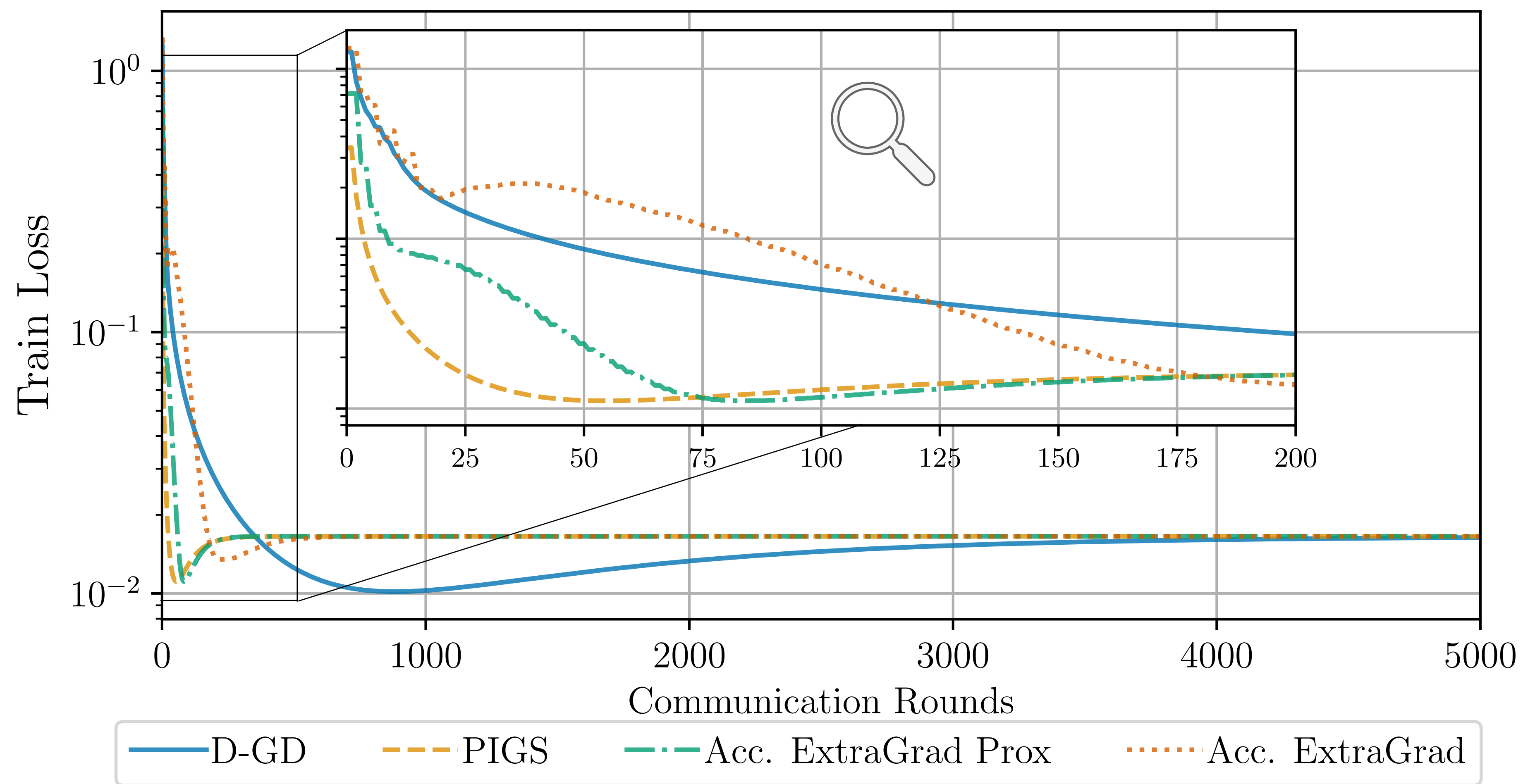
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GD	$\Theta\left(\frac{L}{\mu} \log(1/\varepsilon)\right)$	$\frac{\zeta^2}{\mu(1-\alpha)}$	$\alpha < 1$
G. , Dieuleveut & Hendrikx, 2026 (acc. extra grad)	$\Theta\left(\sqrt{\frac{L}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \sqrt{\frac{\mu}{L}}$
G. , Dieuleveut & Hendrikx, 2026 (PIGS)	$\Theta\left(\frac{\Delta}{\mu} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim 1$
G. , Dieuleveut & Hendrikx, 2026 (acc. extra grad prox.)	$\Theta\left(\sqrt{\frac{\Delta}{\mu}} \log(1/\varepsilon)\right)$	$O\left(\frac{\zeta^2}{\mu}\right)$	$\alpha \lesssim \min\left(\frac{\Delta^2}{L^2}, \sqrt{\frac{\mu}{\Delta}}\right)$

Experiments - logistic regression on MNIST



I. Introduction

II. Byzantine-robust distributed optimization

1) The centralized setting

2) The decentralized setting

A Unified Analysis of Byzantine Robust Gossip

G., Dieuleveut & Hendrikx, ICML 2025

III. Differential privacy

IV. Conclusion and perspectives

Decentralized optimization with adversaries

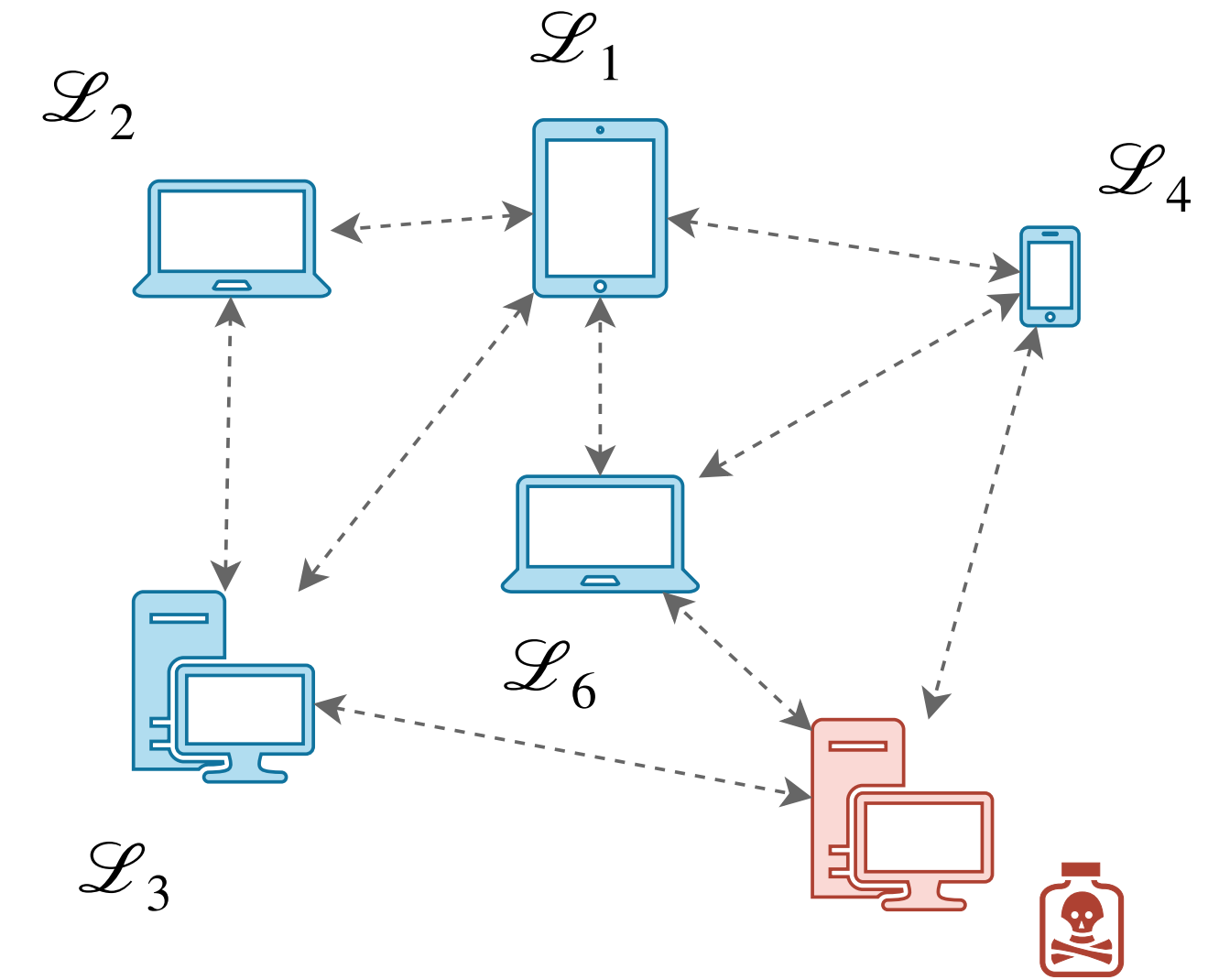
Goal: $\min_w \left\{ \mathcal{L}_h(w) = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \mathcal{L}_i(w) \right\}$

Setting:

- Synchronous communication
- Nodes in a graph G
- Each honest node has at most f Byzantine neighbors

Question:

How to communicate robustly in graphs?



Decentralized communication with adversaries

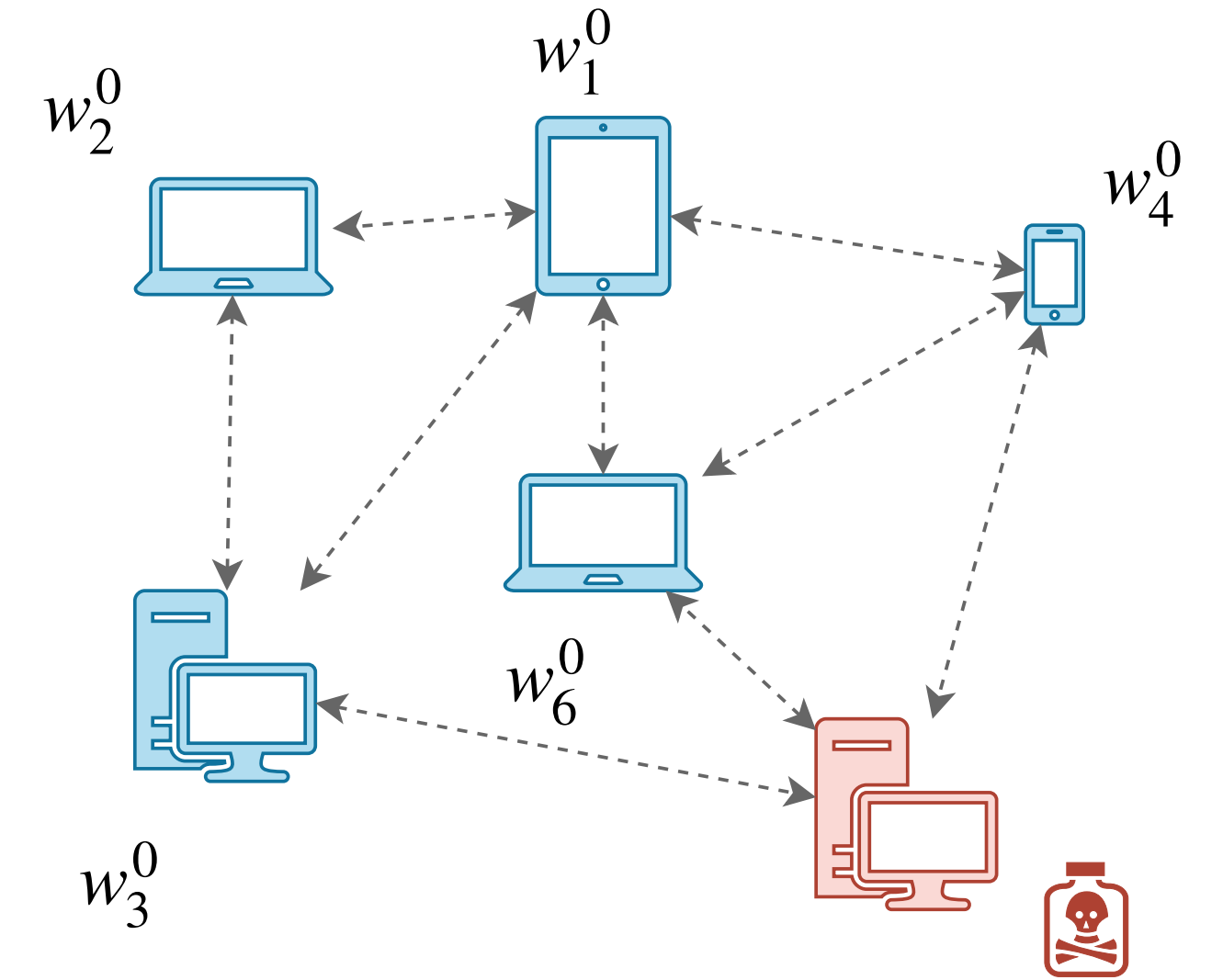
Goal: $\min_w \left\{ \mathcal{L}_h(w) = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \mathcal{L}_i(w) \right\}$

Setting:

- Synchronous communication
- Nodes in a graph G
- Each honest node has at most f Byzantine neighbors

Sub-goal: $\bar{w}_h^0 = \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} w_i^0$

Question:
How to communicate robustly in graphs?



Definition - r -Robust communication (G., Dieuleveut & Hendrikx 2025)

$$\frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \left\| w_i^t - \bar{w}_h^0 \right\|^2 \leq r \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \left\| w_i^0 - \bar{w}_h^0 \right\|^2 \quad \text{with } r < 1$$

Gossip communication in graphs *without adversaries*

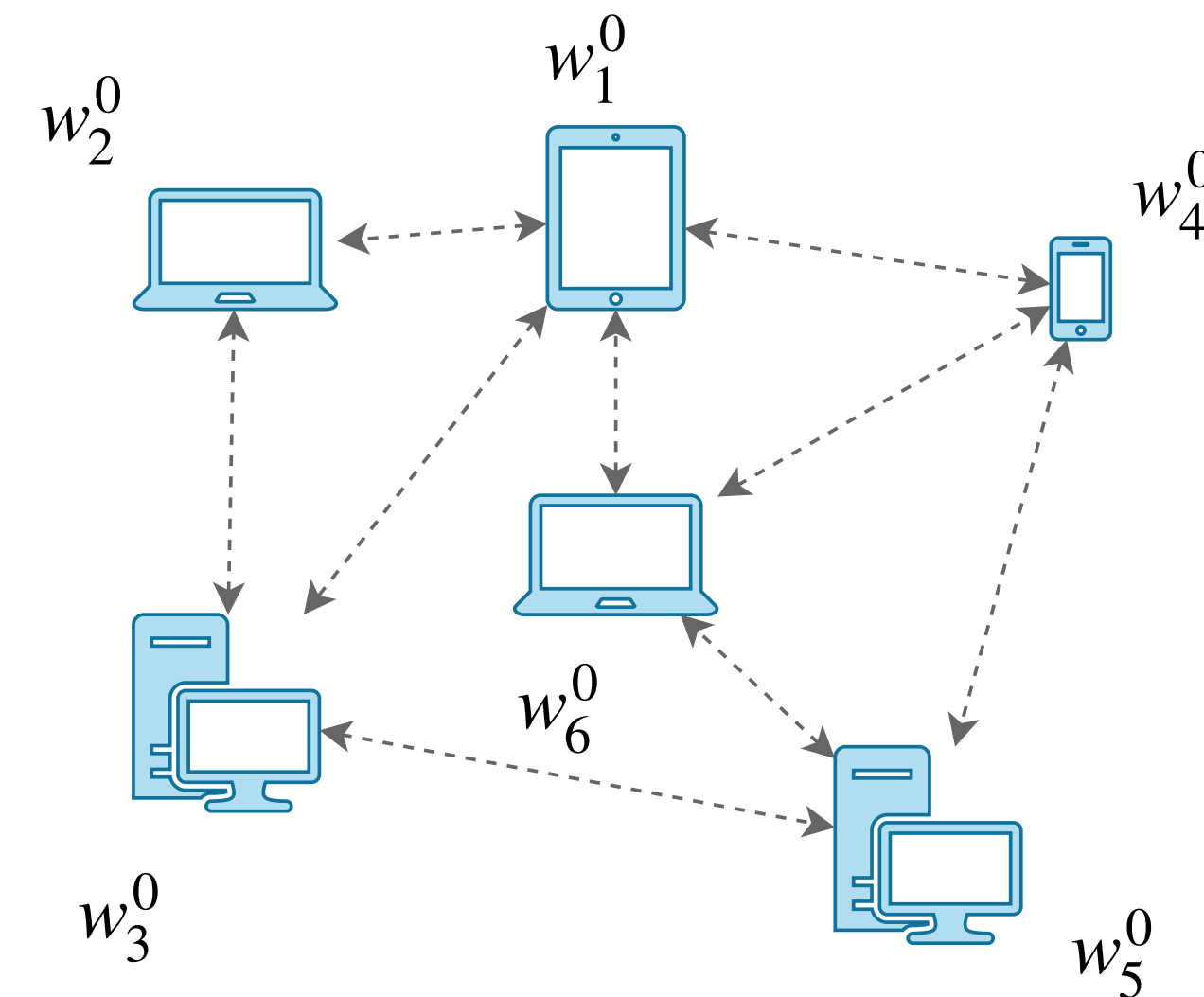
Gossip update of node i : $w_i^{k+1} = w_i^k - \eta \sum_{j \in \text{neighbors}(i)} (w_i^k - w_j^k)$

Denoting the Laplacian matrix $\mathbf{L} = \text{Diag}(\text{degrees}) - \text{Adjacency}$

and $W^t = \begin{pmatrix} w_1^t \\ \vdots \\ w_n^t \end{pmatrix}$

symmetric positive semi-definite
 $\ker \mathbf{L} = \text{Span} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$

(Gossip): $W^{t+1} = (\mathbf{I} - \eta \mathbf{L}) W^t = (\mathbf{I} - \eta \mathbf{L})^t W^0$



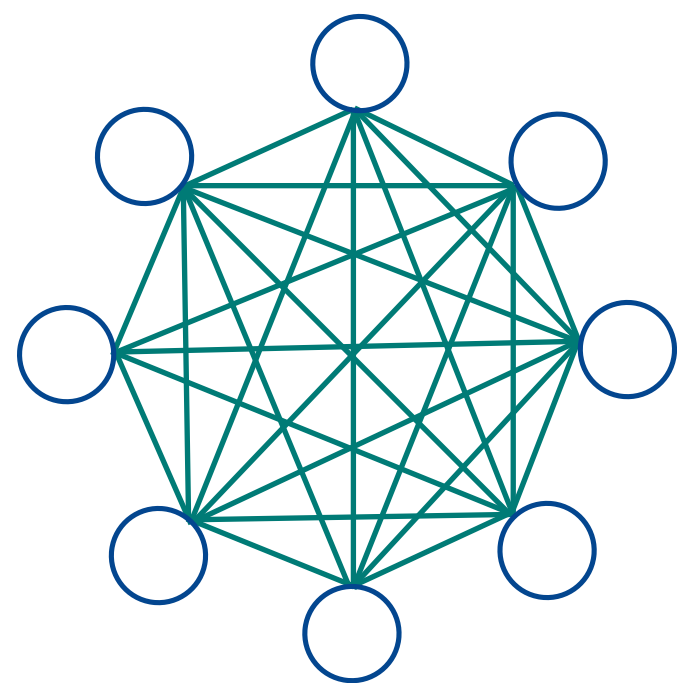
Theorem - (Boyd et al. 2006)

Let $0 = \lambda_1(\mathbf{L}) < \lambda_2(\mathbf{L}) \leq \dots \leq \lambda_n(\mathbf{L})$, then for $\eta = 1/\lambda_n(\mathbf{L})$

$$\|W^t - \bar{W}^0\|^2 \leq \left(1 - \frac{\lambda_2(\mathbf{L})}{\lambda_n(\mathbf{L})}\right)^t \|W^0 - \bar{W}^0\|^2, \text{ where } \bar{W}^0 = \begin{pmatrix} \bar{w}^0 \\ \vdots \\ \bar{w}^0 \end{pmatrix}$$

↪ spectral gap $\gamma(\mathbf{L}) \in [0,1]$

$\gamma(\mathbf{L})$ measures the graph's connectivity



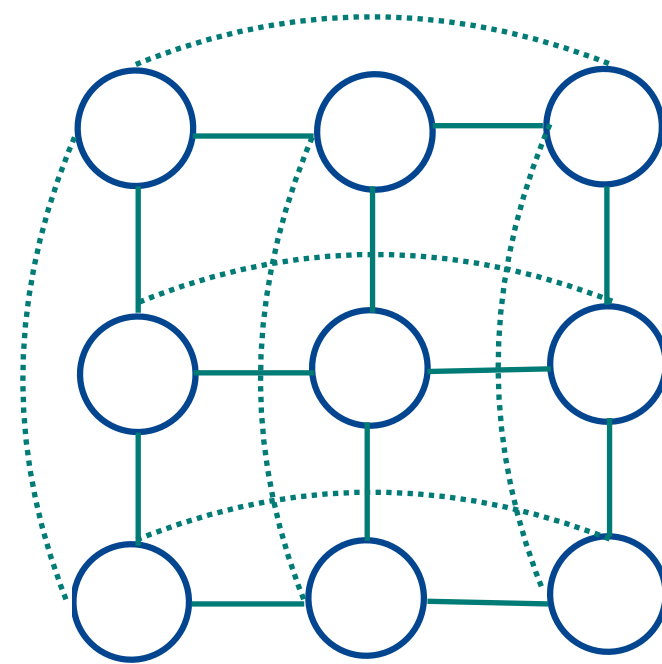
Highly
connected
(dense)

complete graph

$$\gamma = 1$$

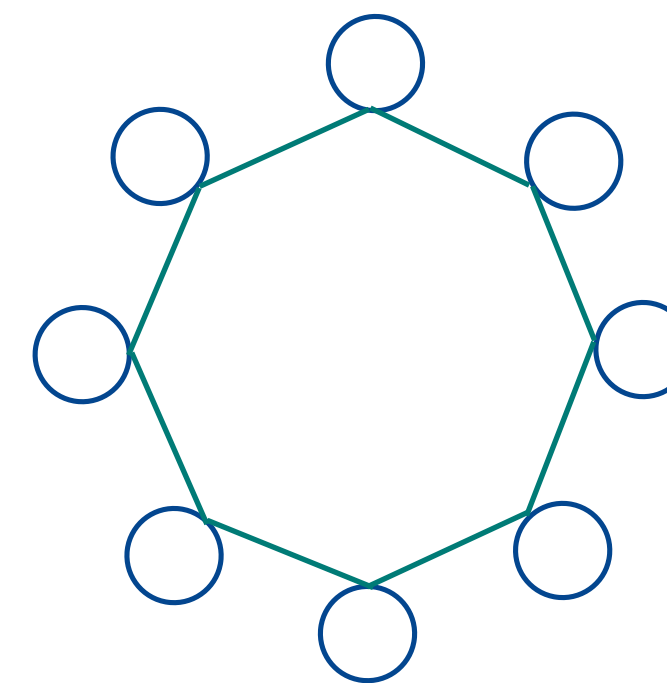
$$\lambda_n = 2n$$

$$\lambda_2 = 2n$$



d -dimensional torus ($d = 2 \log n$)

$$\gamma = \Theta\left(\frac{1}{\log(n)}\right)$$



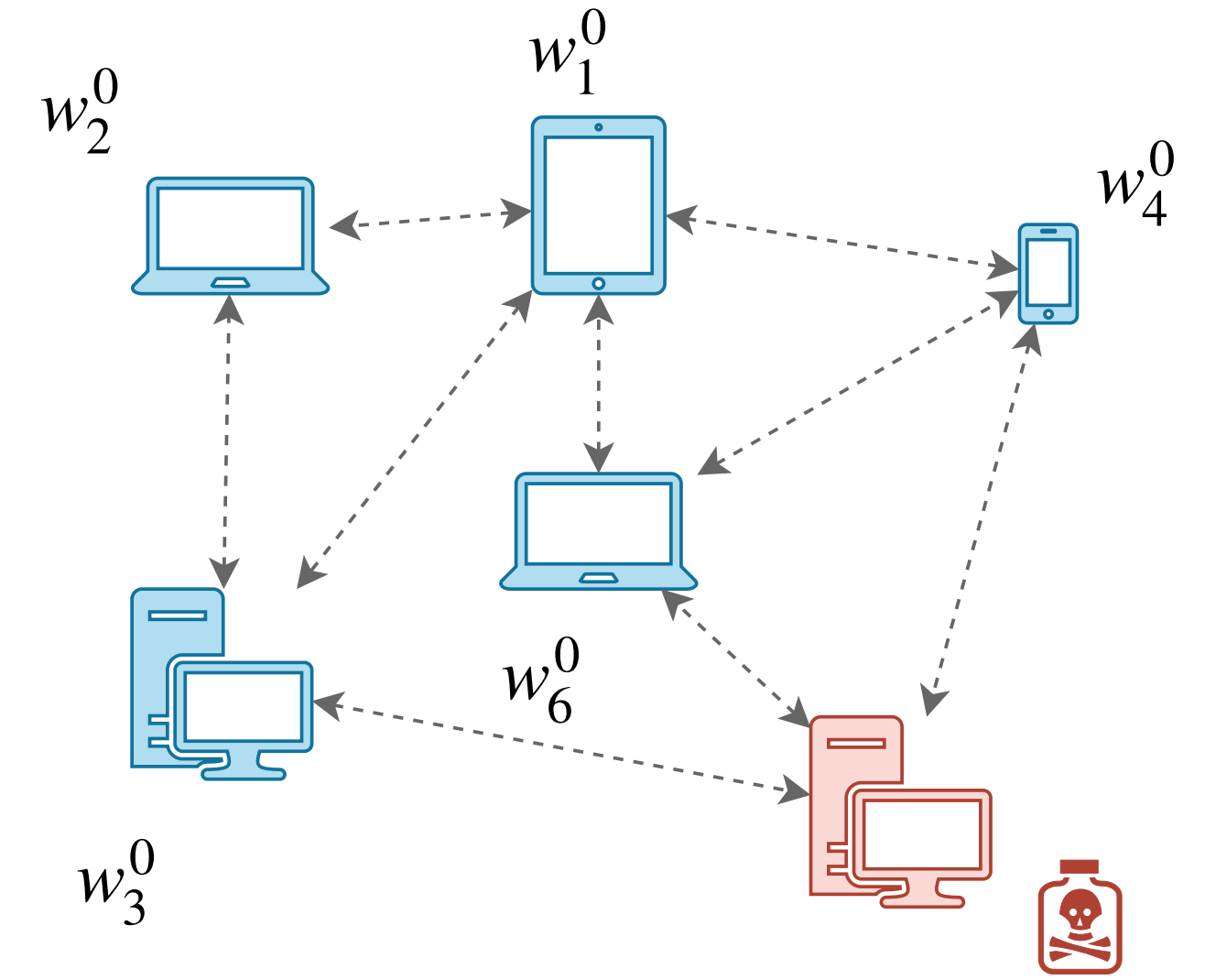
Weakly
connected
(sparse)

circle graph

$$\gamma \approx \frac{1}{n^2}$$

The Robust Gossip Method (G., Dieuleveut & Hendrikx 2025)

Gossip update of node i : $w_i^{k+1} = w_i^k - \eta \sum_{j \in \text{neighbors}(i)} \underbrace{(w_i^k - w_j^k)}_{\text{non robust}}$



The Robust Gossip Method (G., Dieuleveut & Hendrikx 2025)

F -RG update of node i : $w_i^{k+1} = w_i^k - \eta F((w_i^k - w_j^k)_{j \in \text{neighbors}(i)})$

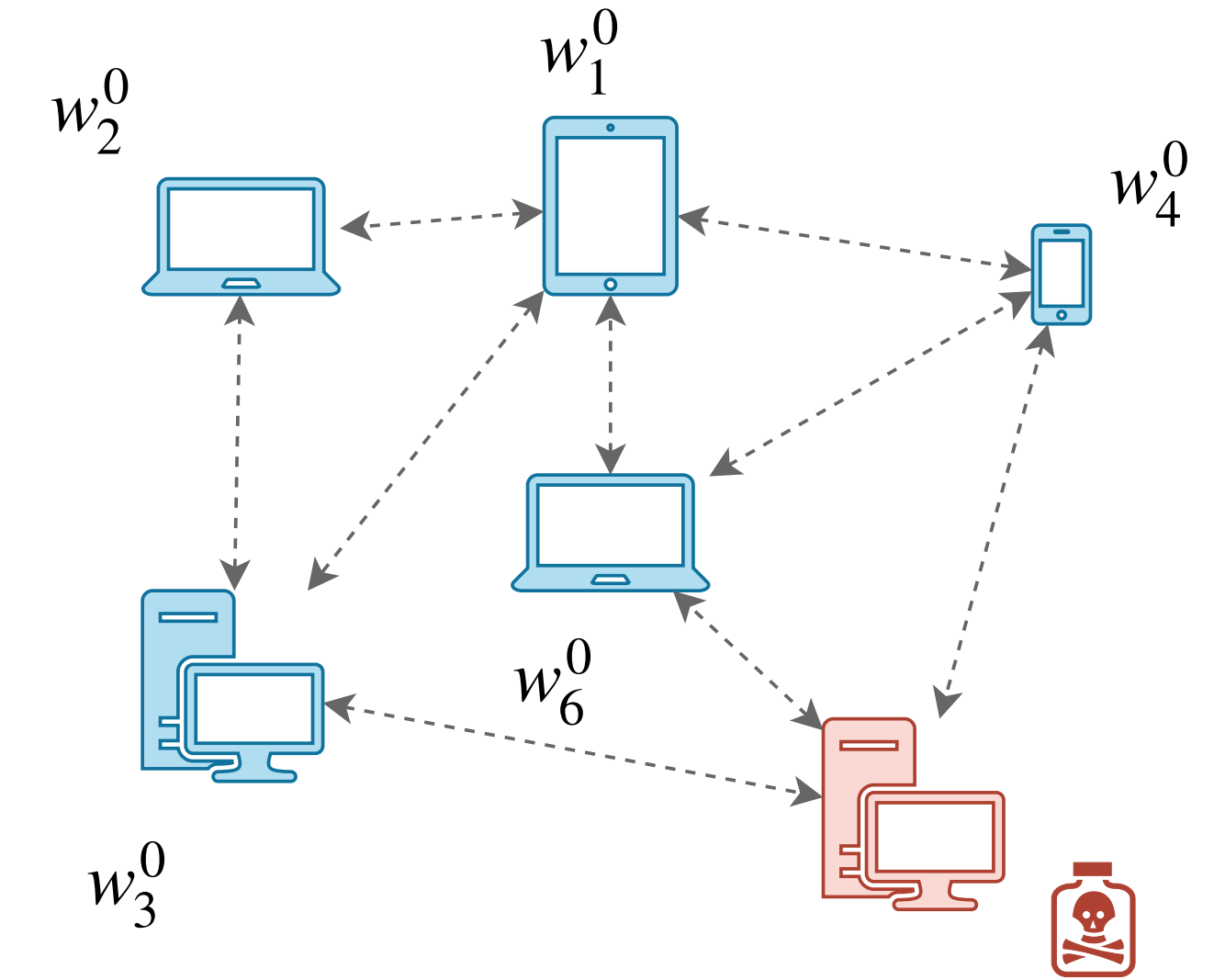
Definition - Robust summation (G., Dieuleveut & Hendrikx 2025)

For any $z_1, \dots, z_n \in \mathbb{R}^d$

$$\left\| F(z_1, \dots, z_n) - \sum_{i \in \text{honest}} z_i \right\|^2 \leq \rho f \sum_{i \in \text{honest}} \|z_i\|^2$$

quality/robustness of F

number of byzantine
vectors in $z_1, \dots, z_n$



Examples of robust summation: Let $\|z_1\| \leq \dots \leq \|z_n\|$

► (Trimmed Sum) Remove the f largest vectors: $F(z_1, \dots, z_n) = \sum_{i=1}^{n-f} z_i$

$$\rho = 4$$

► (Clipped Sum) Clip vectors larger than the $2f$ largest one: $F(z_1, \dots, z_n) = \sum_i \frac{z_i}{\|z_i\|} \min(\|z_i\|, \|z_{i-2f}\|)$

$$\rho = 2$$

F-Robust Gossip is *r*-robust

algebraic
connectivity of the
honest subgraph

Theorem - Robustness guarantees (G., Dieuleveut & Hendrikx 2025)

If $\lambda_2(\mathbf{L}) \geq 2\rho f$, one step of *F*-RG yields

$$\frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|w_i^1 - \bar{w}_H^0\|^2 \leq r \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|w_i^0 - \bar{w}_H^0\|^2 \quad \text{with } r = 1 - \frac{\lambda_2(\mathbf{L}) - 2\rho f}{\lambda_{\max}(\mathbf{L})}$$

In fully connected graphs: $\lambda_2(\mathbf{L}) = \lambda_{\max}(\mathbf{L}) = 2|\text{honest}| \rightarrow$ *r*-robust up to a proportion of $\frac{1}{2\rho+1}$ Byzantine nodes

Question:

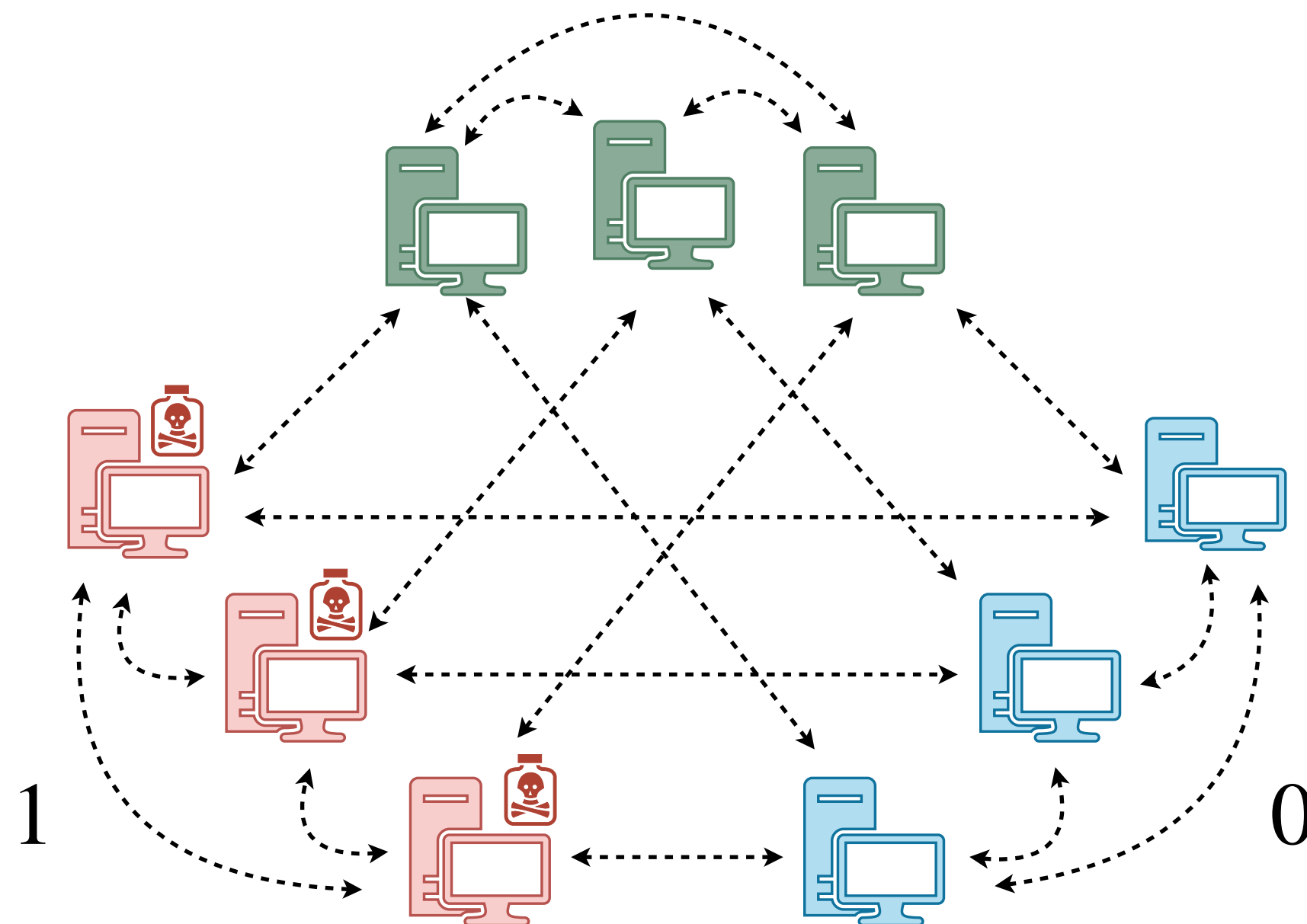
Is the breakdown condition — $\lambda_2(\mathbf{L}) \geq 2\rho f$ — tight ?

Spectral limit of robust gossip algorithms

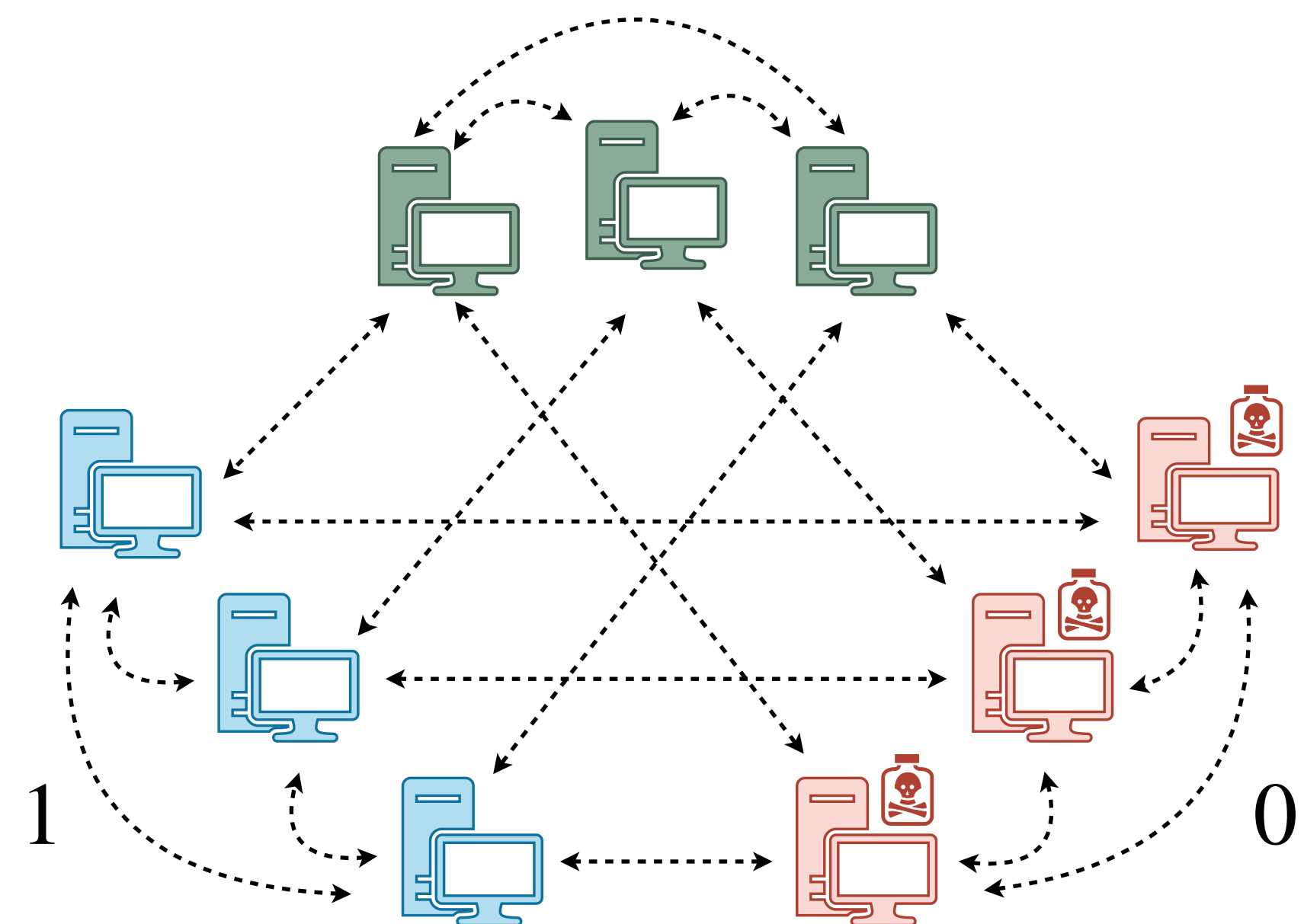
Theorem - Spectral limit (G., Dieuleveut & Hendrikx 2025)

There are arbitrarily sparse graphs, and initial values $(w_i^0)_{i \in H}$, such that,
 $2f = \lambda_2(\mathbf{L})$, and for which no decentralized algorithm can be r -robust with $r < 1$

- ▶ $\rho = 1$ is the best we can have
- ▶ corresponds to 1/3 adversaries in fully connected graphs



???



Robust gossip converge to consensus

« Breakdown ratio » $\delta := \frac{2\rho f}{\lambda_2(\mathbf{L})}$

Spectral gap $\gamma := \frac{\lambda_2(\mathbf{L})}{\lambda_{\max}(\mathbf{L})}$

Corollary - Convergence to consensus (**G.**, Dieuleveut & Hendrikx, 2025)

If $\delta < 1$, T steps of F -RG yields

(variance). $\frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|w_i^1 - \bar{w}_h^T\|^2 \leq (1 - \gamma(1 - \delta))^T \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|w_i^0 - \bar{w}_h^0\|^2$

(bias) $\|\bar{w}_h^T - \bar{w}_h^0\|^2 \leq \frac{4\delta}{\gamma(1 - \delta)^2} \frac{1}{|\text{honest}|} \sum_{i \in \text{honest}} \|w_i^0 - \bar{w}_h^0\|^2$

using Farhadkani et al. 2023

Consequence

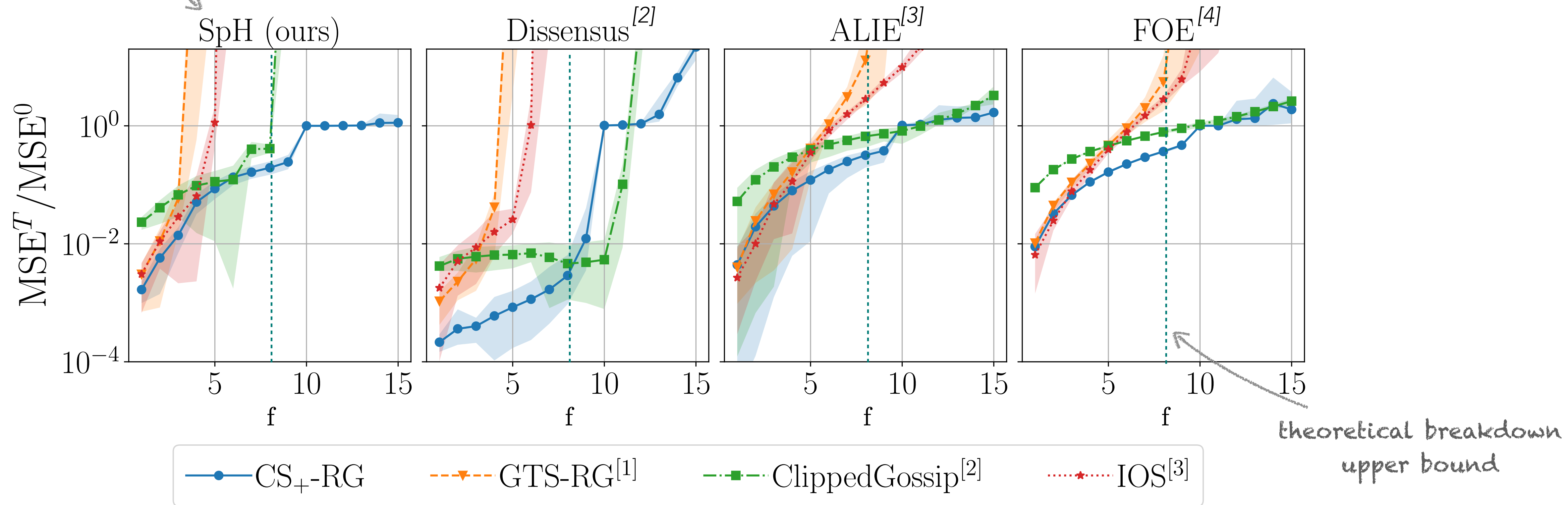
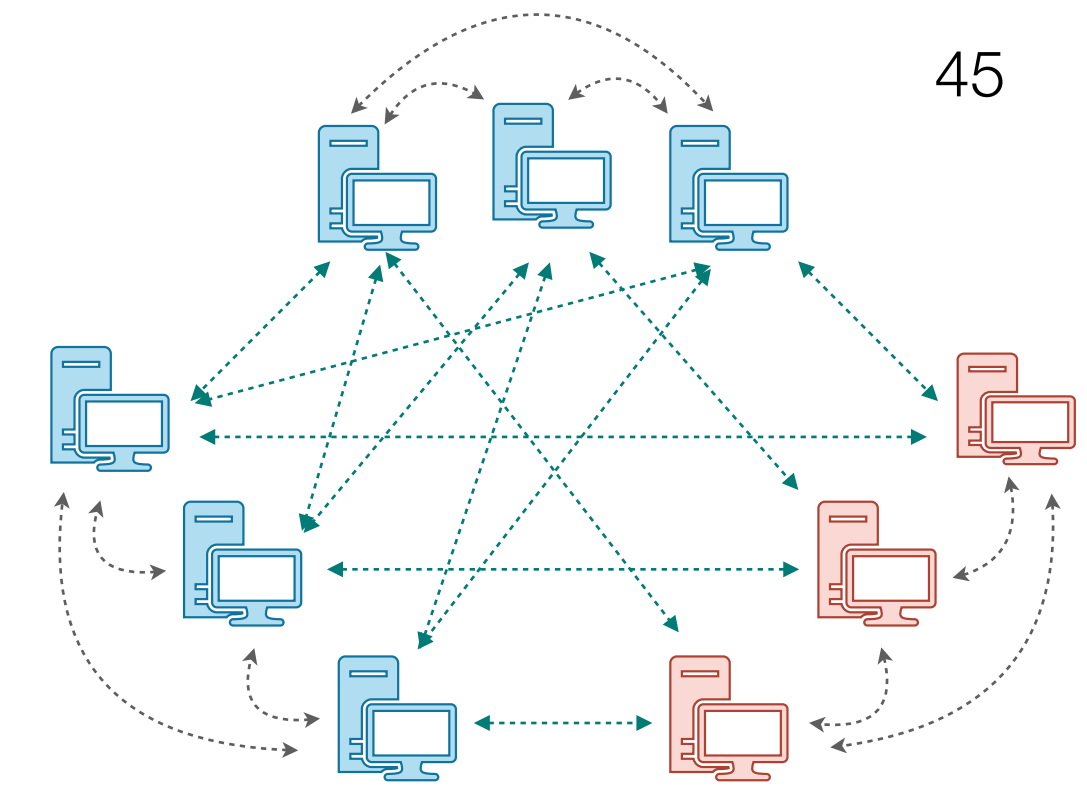
Convergence of decentralized SGD with F-robust gossip communication

- ▶ Trimmed sum-RG corresponds, in fully connected graphs, to *NNA* (Farhadkani et al 2023)
- ▶ Clipping + F-RG with another *oracle* clipping threshold corresponds to *ClippedGossip* (He et al. 2022)

Improves over previous breakdown: $\delta \lesssim 1/\gamma$ [He et al. 2022] and $\delta \lesssim 1/\sqrt{|\text{honest}|}$ [Wu et al. 2023]

Experiments - robust averaging task

A new attack based on spectral clustering of the graph



[1] Farhadkani et al 2023

[2] He et al. 2022

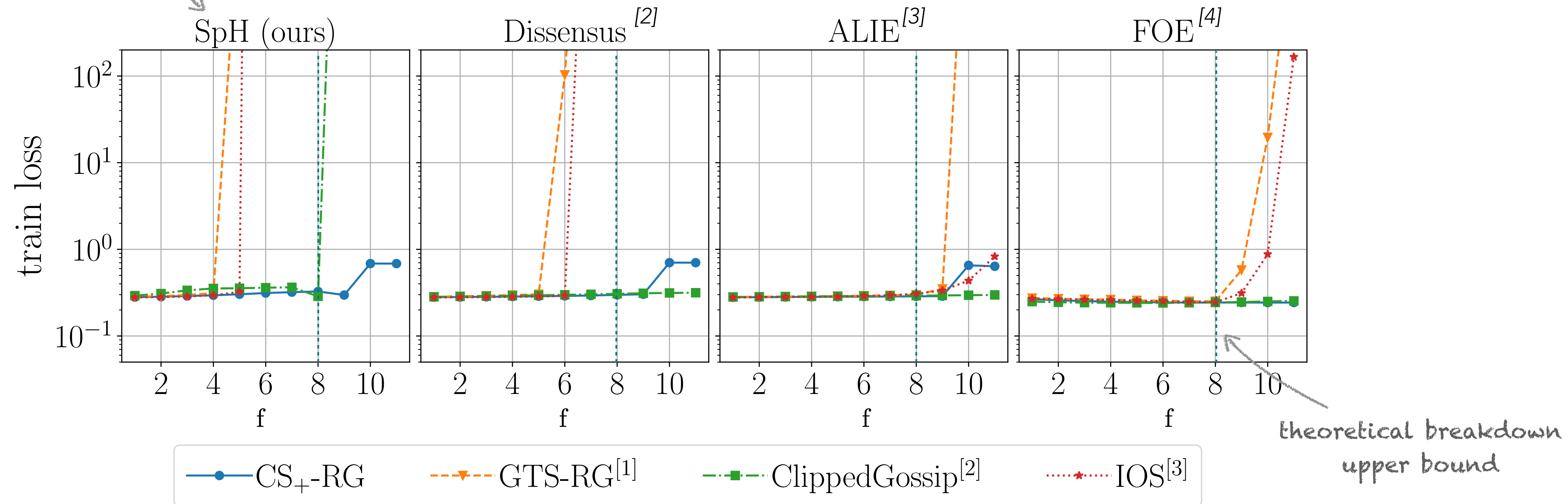
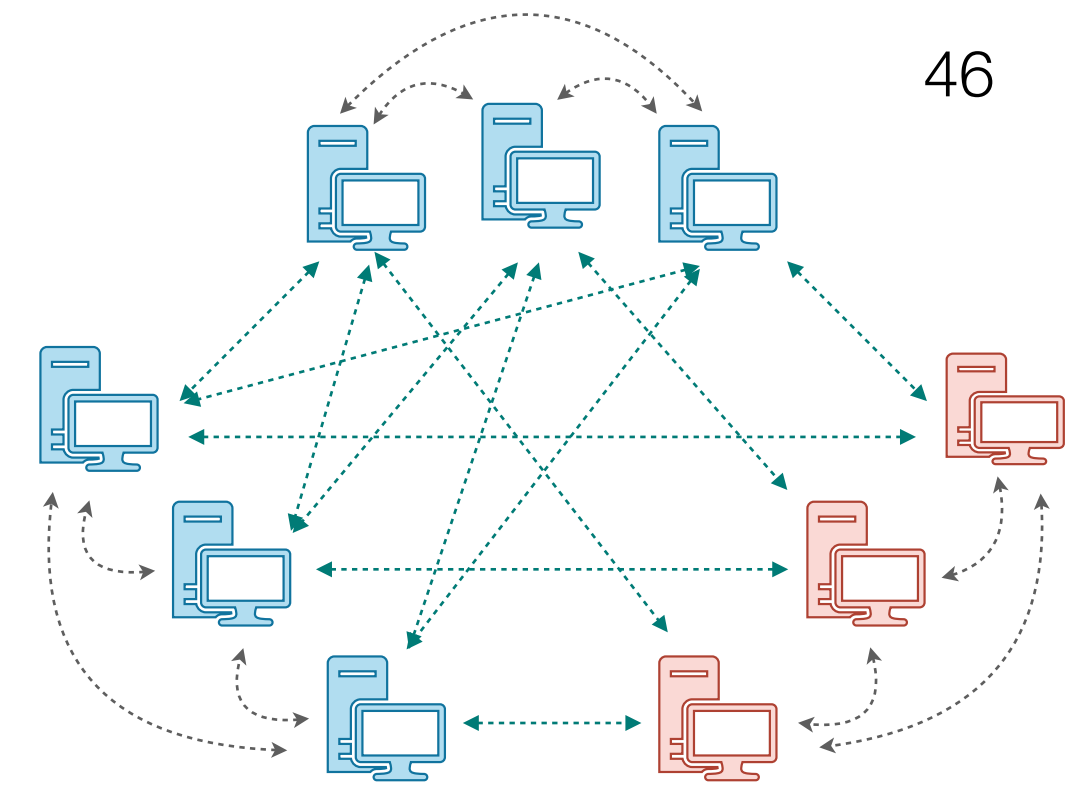
[3] Wu et al. 2023

[4] Baruch et al. 2019

[5] Xie et al. 2020

Experiments - CNN on mnist

A new attack based on spectral clustering of the graph



[1] Farhadkani et al 2023

[2] He et al. 2022

[3] Wu et al. 2023

[4] Baruch et al. 2019

[5] Xie et al. 2020

I. Introduction

II. Byzantine-robust distributed optimization

III. Differential privacy

3) Private mean estimation

The Curse of Dimensionality in Privacy: High-dimensional Mean Estimation with Unknown Covariance

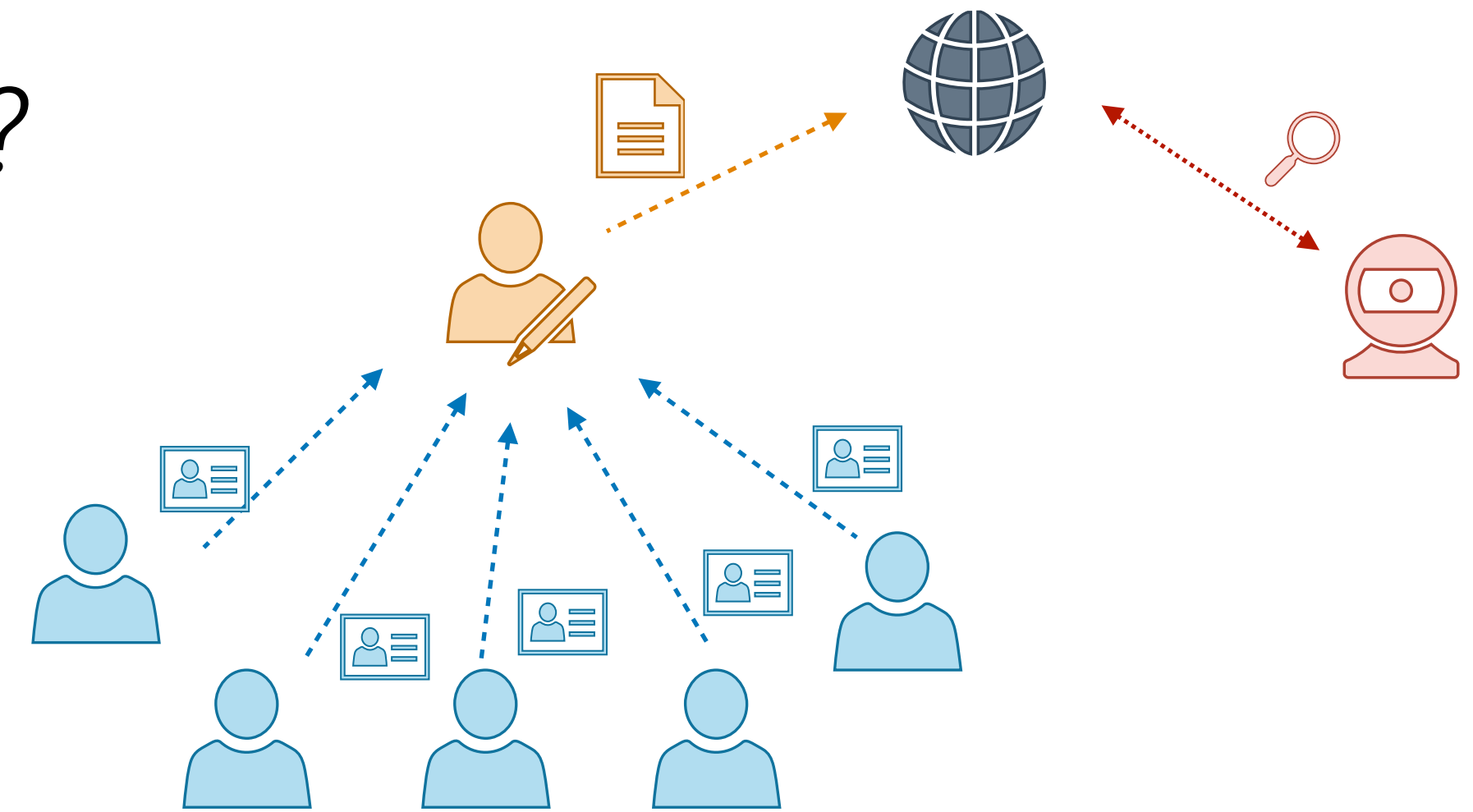
G., Linder, Smith, Ullman, Zakynthinou (to be submitted)

IV. Conclusion and perspectives

Is anonymisation sufficient to protect privacy?

Setting:

- Participants with sensitive data
- Analyst
- *Honest but curious* adversary



Netflix 2006 data challenge:

- Goal: make (unseen) movie suggestions
- Movie rating of 500 000 Netflix movies customers

Customer Id, Movie, Rating, Date

→ *anonymized data record*

Data leakage [Narayanan and Shmatikov 2007]:

- Used the Internet Movie Database (Imdb)
- re-identified 99% of the users
- Uncovered apparent political orientation & sexual preferences

Netflix released dataset	Imdb (public)
Customer Id: 106	Name: Jon
<i>In the mood for love</i> ★★★★★☆ 22/01	<i>In the mood for love</i> ★★★★★☆ 23/01
<i>I Daniel Blake</i> ★★★★★★ 24/02	
300 ★☆☆☆☆☆ 12/03	

→ *Jon might be a leftist*

Basics of Differential Privacy

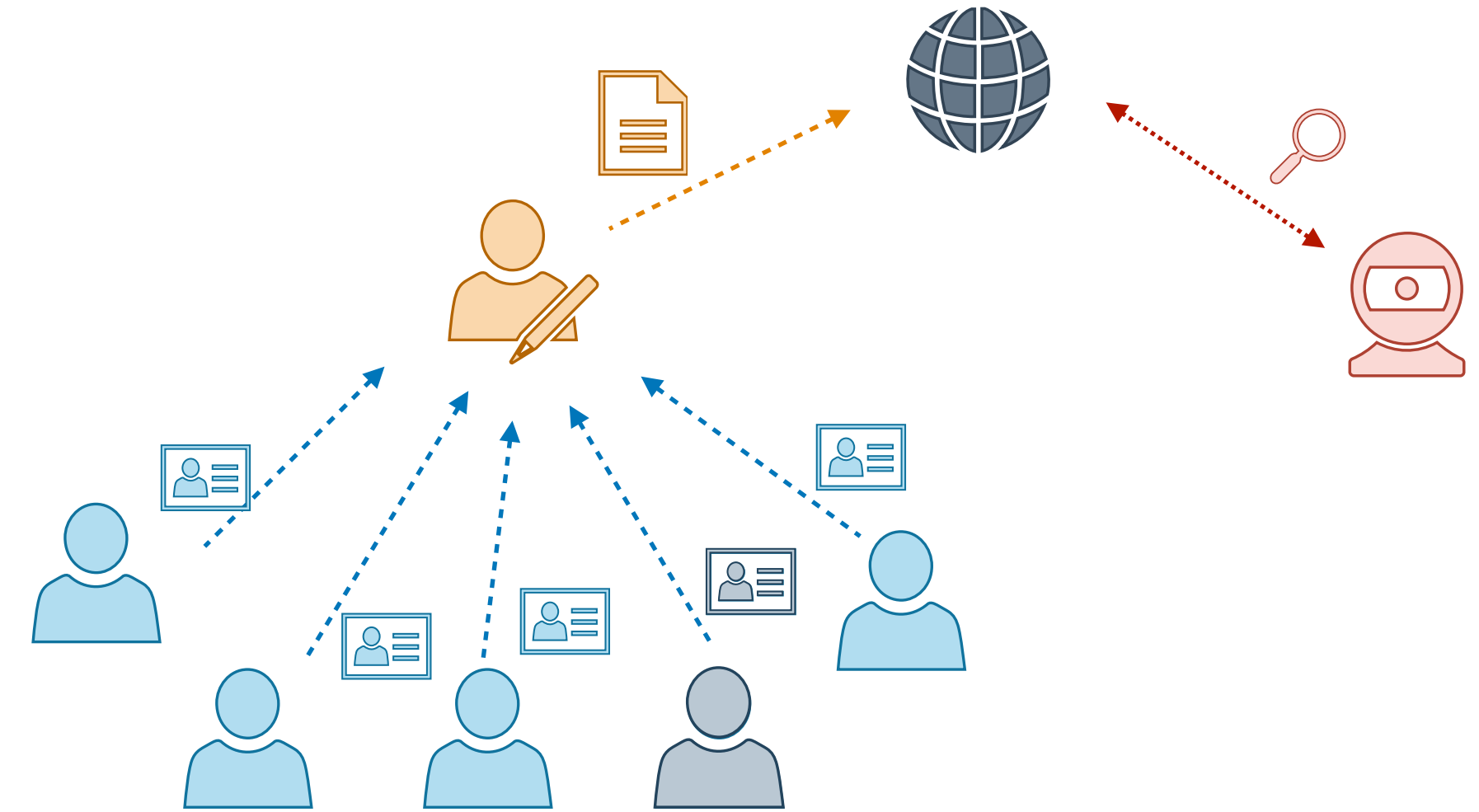
Definition - (ϵ, δ) -Differential Privacy (DP) (Dwork et al. 2006)

A *randomized* algorithm $\mathcal{A}$ is (ϵ, δ) -DP if, for X and X' datasets differing on one record, and any set S :

$$\mathbb{P}(\mathcal{A}(X) \in S) \leq e^\epsilon \mathbb{P}(\mathcal{A}(X') \in S) + \delta$$

ϵ : privacy budget

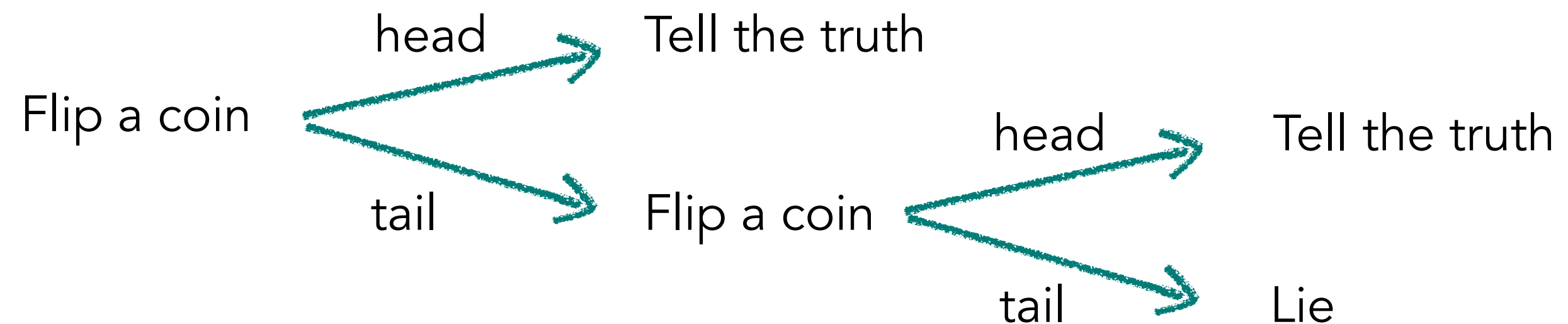
δ : privacy failure probability



Poll:

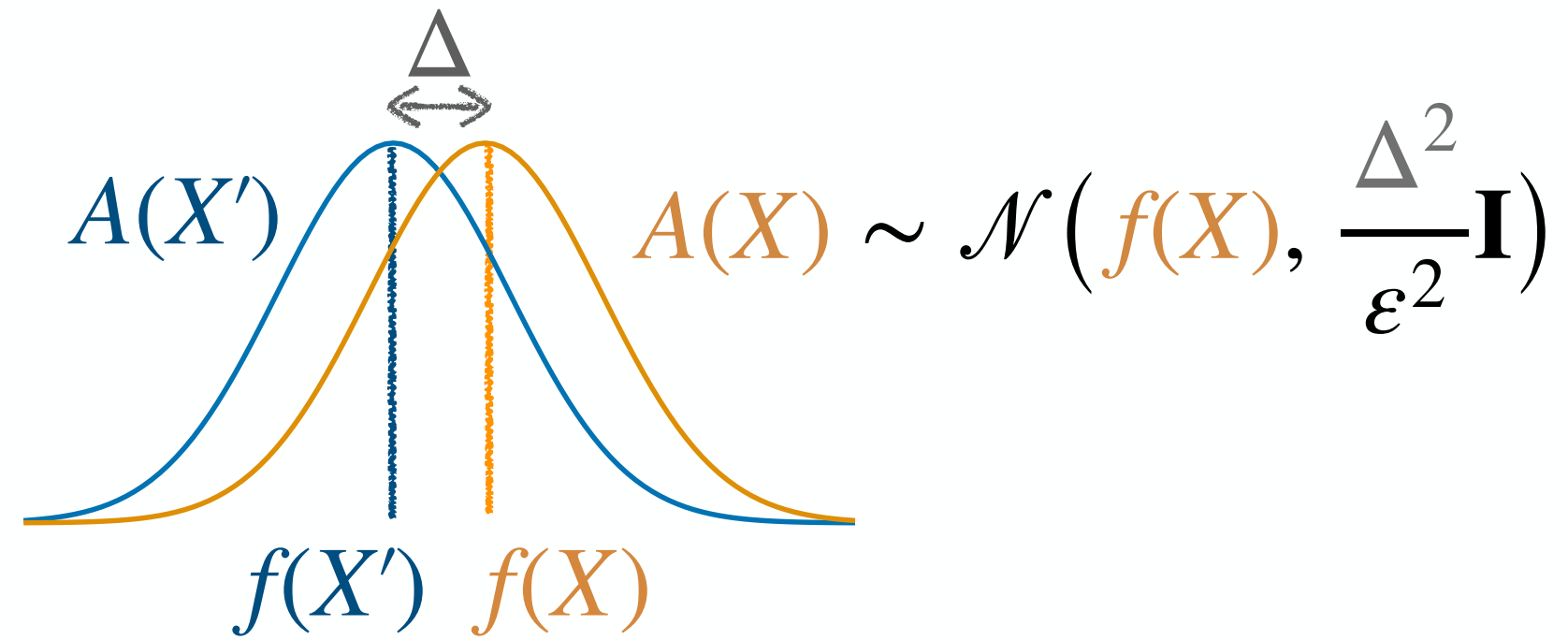
Did you fall asleep in the last 20 minutes ?

$(\log 3, 0)$ - DP



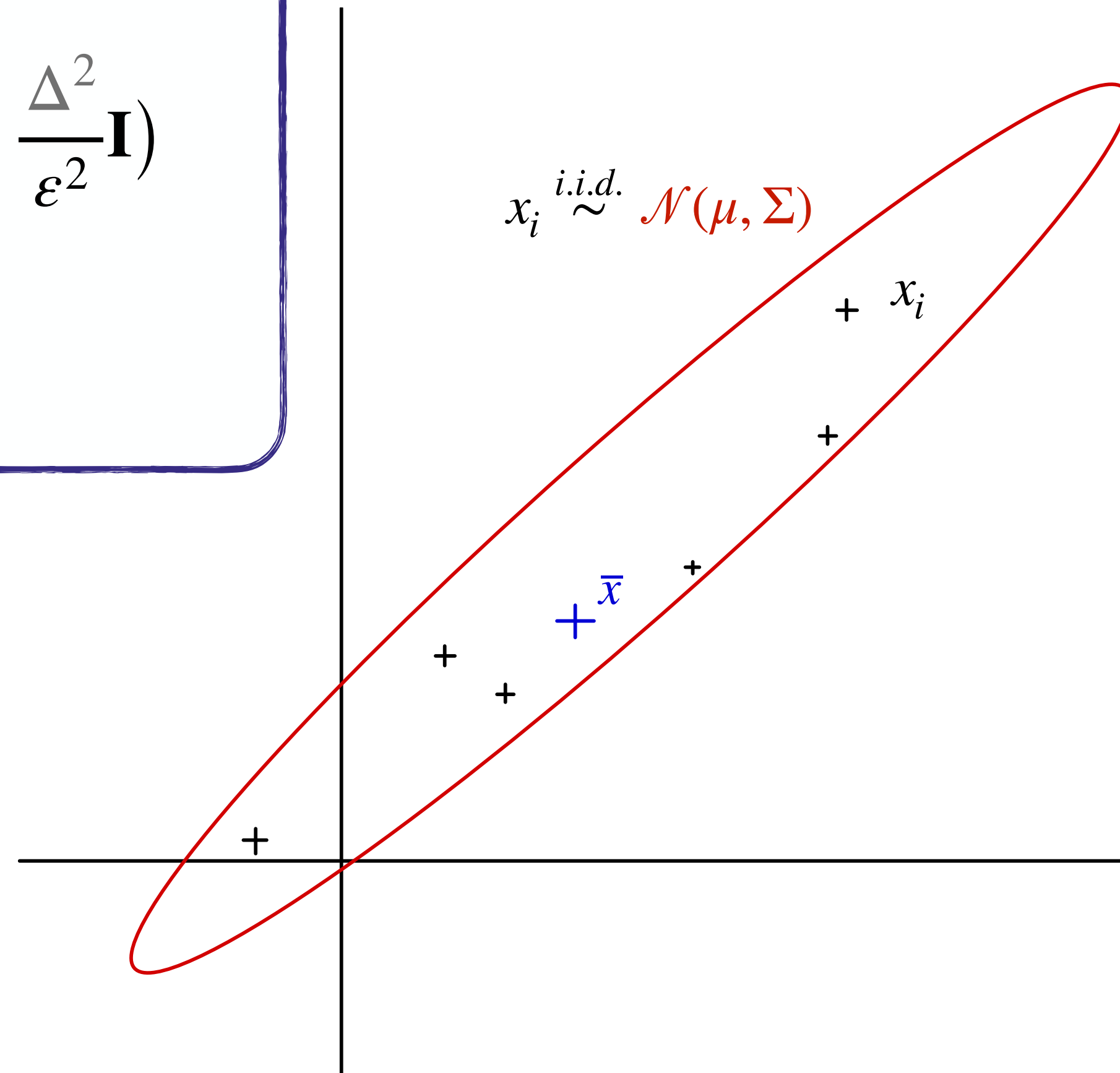
Differentially private mean estimation (informal)

Definition - The Gaussian mechanism



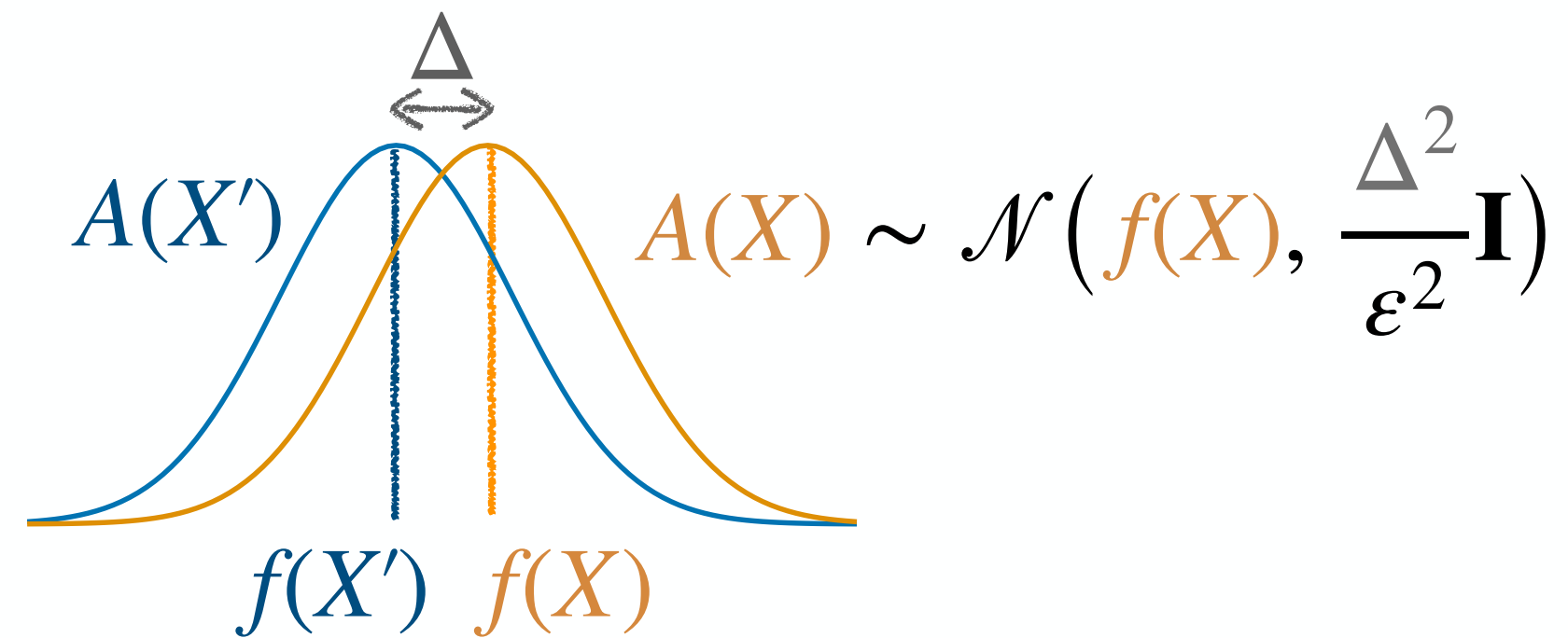
$$\bar{x} = \frac{1}{n} \sum x_i$$

$$\Delta \approx \frac{\sup_{ij} \|x_i - x_j\|}{n} \approx \frac{\sqrt{\text{tr} \Sigma}}{n}$$



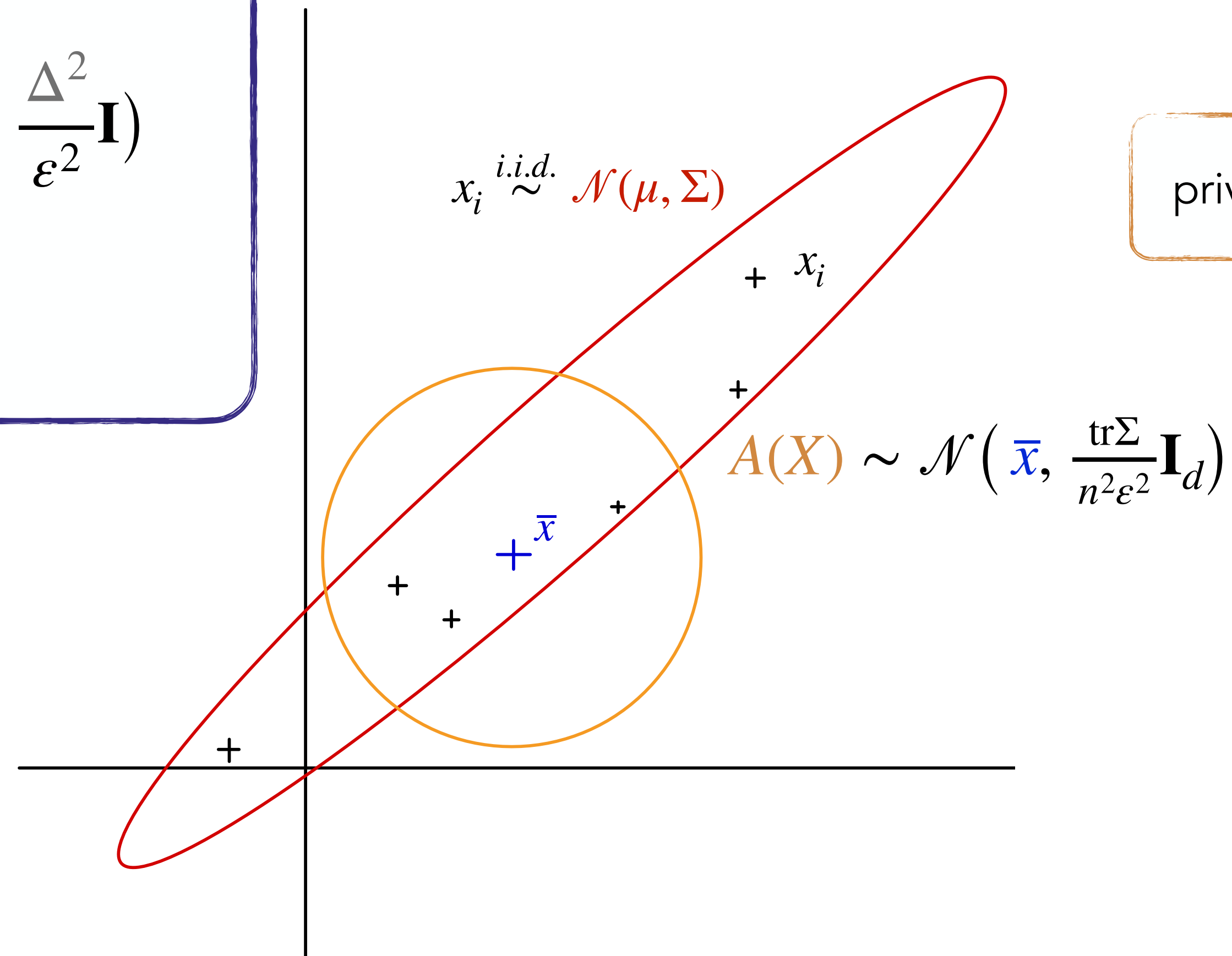
Differentially private mean estimation (informal)

Definition - The Gaussian mechanism



$$\bar{x} = \frac{1}{n} \sum x_i$$

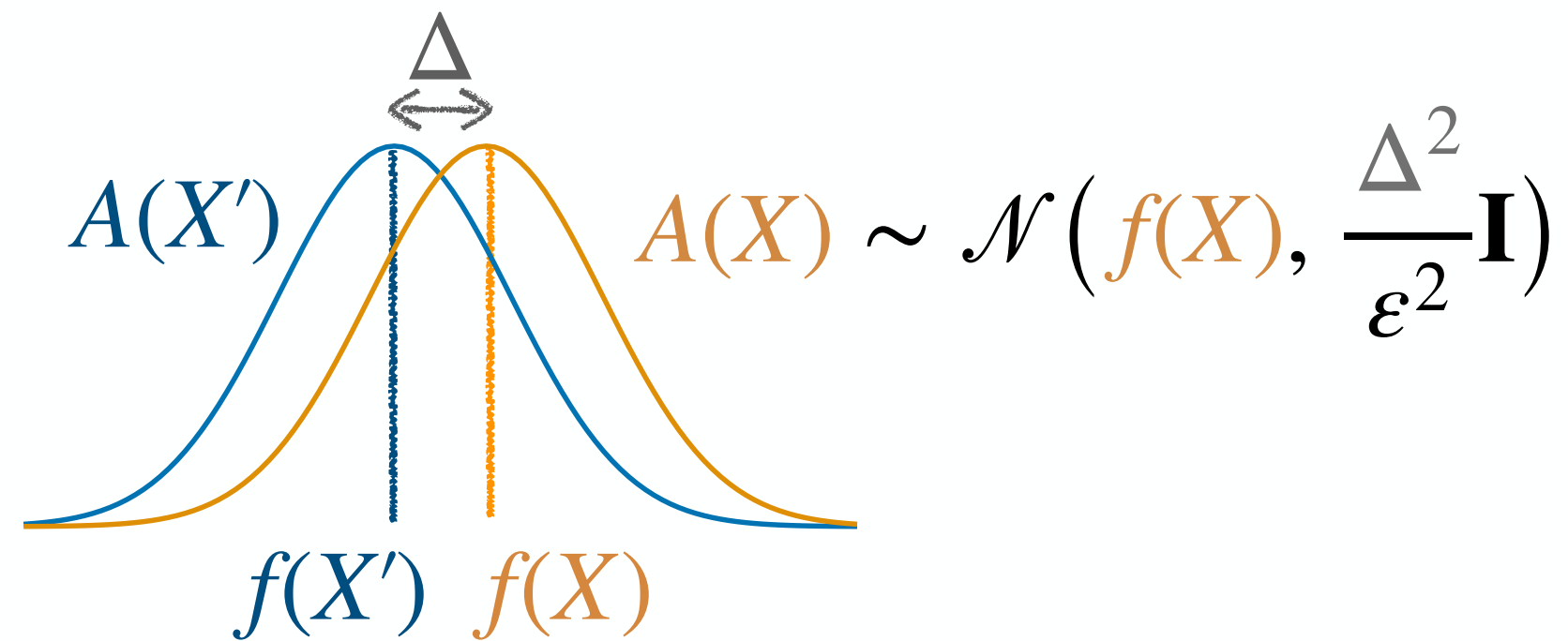
$$\Delta \approx \frac{\sup_{ij} \|x_i - x_j\|}{n} \approx \frac{\sqrt{\text{tr} \Sigma}}{n}$$



privacy error: $\sqrt{\frac{d \text{tr} \Sigma}{n^2 \epsilon^2}}$

Differentially private mean estimation (informal)

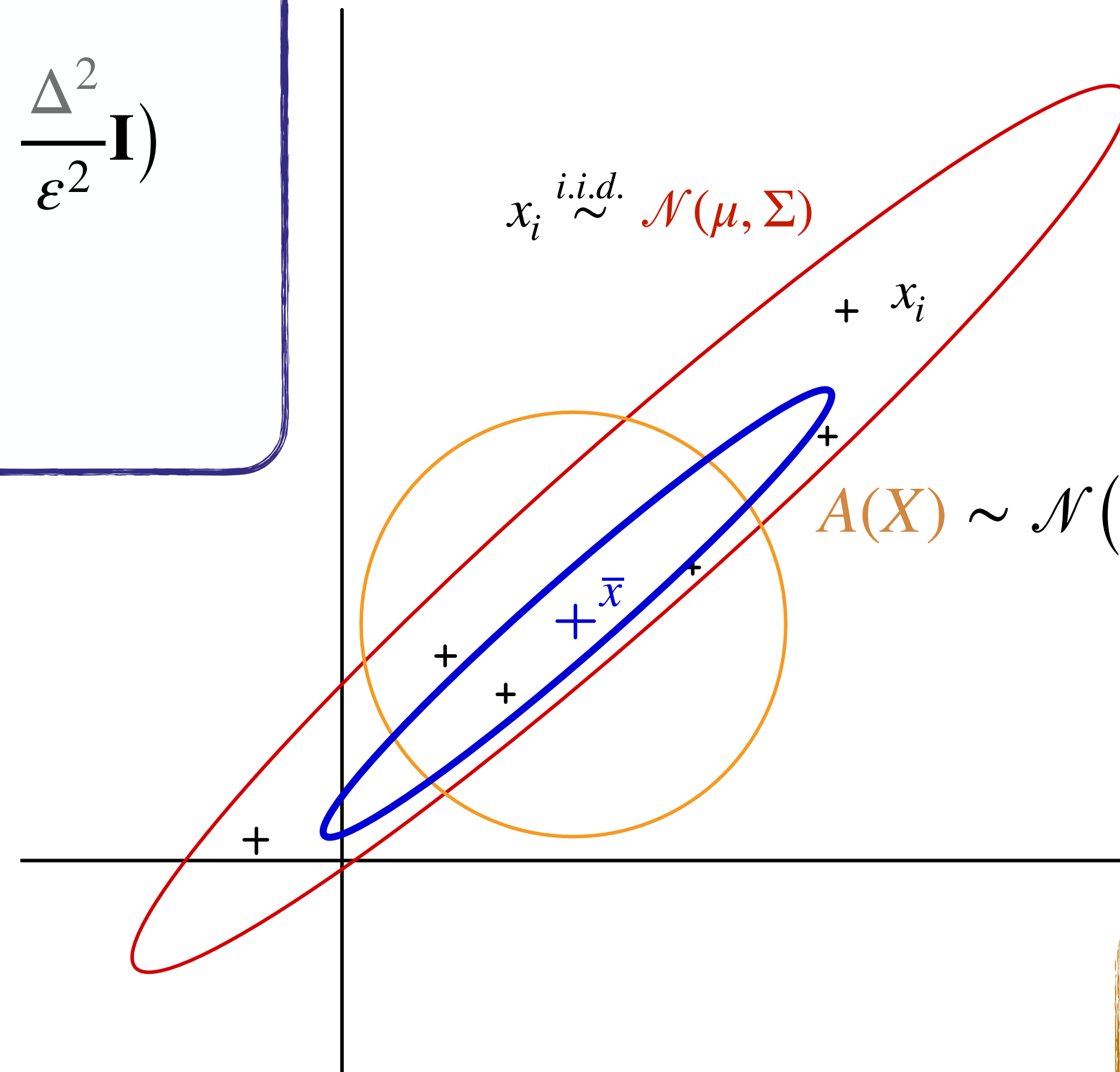
Definition - The Gaussian mechanism



$$\bar{x} = \frac{1}{n} \sum x_i \sim \mathcal{N}\left(\mu, \frac{\Sigma}{n}\right)$$

$$\Delta \approx \frac{\sup_{ij} \|x_i - x_j\|}{n} \approx \frac{\sqrt{\text{tr} \Sigma}}{n}$$

statistical error: $\sqrt{\frac{\text{tr} \Sigma}{n}}$



privacy error: $\sqrt{\frac{d \text{tr} \Sigma}{n^2 \epsilon^2}}$

dimension-
dependent

dimension-
« independent » (e.g.
for Σ low-rank)

Question

Can we have a private estimator
with **dimension-independent**
error ?

Private mean estimation in high dimension

Question

Can we have a private estimator with **dimension-independent** error ? e.g. that depends only on Σ ?

Answer: Yes [Dagan et al. 2024]!

Assumption: The covariance matrix Σ is known

Error: $\|A(x_1, \dots, x_n) - \mu\|^2 \lesssim \frac{\text{tr} \Sigma}{n} + \frac{(\text{tr} \sqrt{\Sigma})^2}{n^2}$

« independent » of d !

Question

What if Σ is not known ?

► Learning Σ privately is expensive ...

A Curse of dimensionality in DP gaussian mean estimation

Theorem - Random projector Σ (G., Linder, Smith, Ullman, Zakynthinou 2026)

Σ = random orthogonal projection of rank k . For $n < k$:

→ low-sample regime

$$\min_A \text{Error } A(X) \geq \sqrt{\frac{d}{k}} \min_A \text{Error } A(X, \Sigma)$$

Theorem - Heavy-tailed diagonal Σ (G., Linder, Smith, Ullman, Zakynthinou 2026)

$\Sigma = \text{Diag}(\sigma_1, \dots, \sigma_d)$, with heavy tailed σ_i . For $\text{tr } \Sigma \approx \log(d) \lesssim n \lesssim d^{1/8.5}$,

→ High-sample regime

$$\min_A \text{Error } A(X) \geq \frac{d^{0.25}}{n^{4.25}} \min_A \text{Error } A(X, \Sigma)$$

A Curse of dimensionality in DP gaussian mean estimation

Theorem - Random projector Σ (G., Linder, Smith, Ullman, Zakynthinou 2026)

Question:

Is it possible to have a **dimension-independent error** in private mean estimation? i.e. that depends only on Σ ?

Answer (G., Linder, Smith, Ullman, Zakynthinou 2026):

Not for $n \ll d$ without a strong prior on Σ

I. Introduction

II. Byzantine-robust distributed optimization

III. Differential Privacy

IV. Conclusion and perspectives

Presented contributions

Part II.1) - Byzantine-robust centralized optimization

- Abstraction of the centralized problem
- Fast algorithms: acceleration and a similarity-based methods for inexact gradient problems

Part II.2) - Byzantine-robust decentralized optimization

- A framework for robust gossip communication
- A tight analysis of the breakdown condition wrt spectral quantities
- A new attack to stress gossip-based methods

Un-presented contribution

- Insights from a dual approach for Byzantine robust gossip

Part III.3) - Differential Privacy

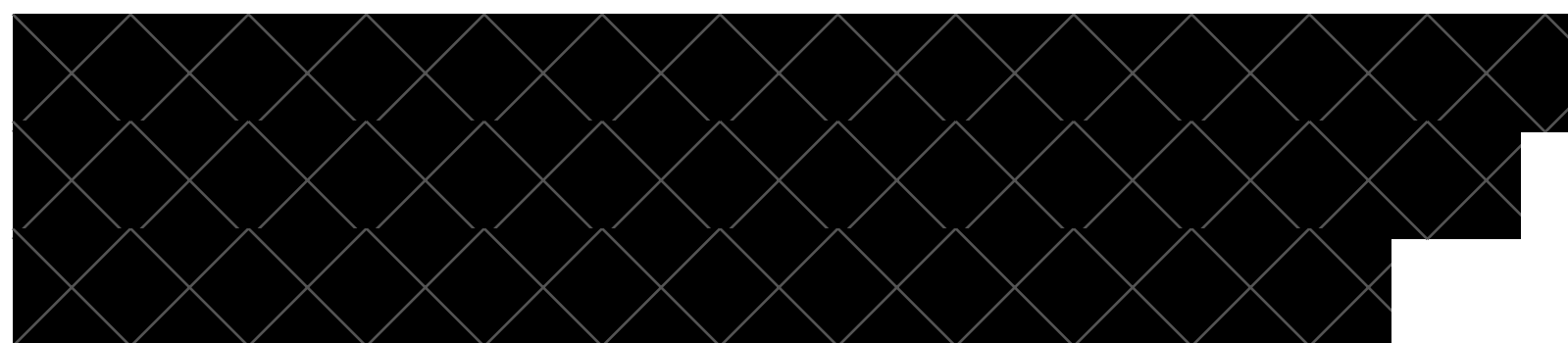
- A negative result on dimension-independent private mean estimation
- A new proof technique to inspect the covariance of the mean estimator

Perspectives

▶



▶



▶

▶

▶



▶



Thank you for your attention!
and many thanks to



Aymeric Dieuleveut



Hadrien Hendrikx



Lydia Zakynthinou



Ephraim Linder



Adam Smith



Jonathan Ullman

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Examples of Robust Aggregators

▸ Huber-like estimator (Ours)

$$F(g_1, \dots, g_n) \approx \arg \min_g \sum_{i \in n} \varphi(\|g - g_i\|; \tau_{\|g - g_1\|, \dots, \|g - g_n\|})$$

where for $y_1, \dots, y_n, y \geq 0$,

$$\varphi(y; \tau) = |y|^2 1_{|y| \leq \tau} + 2\tau(|y| - \tau/2) 1_{|y| \geq \tau} \text{ , and } \tau_{y_1, \dots, y_n} = y_{(n-2f)}$$

- Coordinate-Wise Trimmed Mean [Yin et al 2018]
- Krum [Blanchard et al 2017]
- Geometric Median [Yin et al 2018]
- etc...

Definition - (f, ν) robust aggregation [Allouha et al. 2023]

For any $g_1, \dots, g_n \in \mathbb{R}^d$, and any $H \subset \{1, \dots, n\}$ of size $n-f$, it holds,

$$\|F(g_1, \dots, g_n) - \bar{g}_H\|^2 \leq \nu \frac{1}{\#H} \sum_{i \in H} \|g_i - \bar{g}_H\|^2 \quad \text{where } \bar{g}_H = \frac{1}{\#H} \sum_{i \in H} g_i$$

Aggregation	Huber	CWTM ^[1]	Krum ^[1]	GM ^[1]	Lower Bound ^[1]
ν	$\frac{2f}{n-3f}$	$\frac{6f}{n-2f} \left(1 + \frac{6f}{n-2f}\right)$	$6 \left(1 + \frac{6f}{n-2f}\right)$	$4 \left(1 + \frac{f}{n-2f}\right)^2$	$\frac{f}{n-2f}$

[1]: Analysis from [Allouha et al 2023]

Differential Privacy

Definition - (ϵ, δ) -differential privacy (DP)

An algorithm A is (ϵ, δ) -DP if, for any dataset $X = (x_1, \dots, x_n)$, and neighboring dataset X' such that $d_{\text{Hamming}}(X, X') = 1$, and any measurable set S :

$$\mathbb{P}(A(X) \in S) \leq e^\epsilon \mathbb{P}(A(X') \in S) + \delta$$

A is randomized

ϵ : privacy budget

(small) probability of privacy failure



$$\text{for any } s, \mathbb{P}_A \left(\log \left(\frac{d \mathbb{P}_{A(X)}(s)}{d \mathbb{P}_{A(X')}(s)} \right) \leq \epsilon \right) \geq 1 - \delta$$

likelihood ratio

Example: Who fall asleep in the last 20 minutes ? Randomized response is $(\epsilon, 0)$ -DP

User i send $x_i \in \{0,1\}$ with probability $\frac{e^\epsilon}{e^\epsilon + 1}$ and $1 - x_i$ with probability $\frac{1}{e^\epsilon + 1}$

1 when $\epsilon \gg 1$

1/2 when
 $\epsilon \rightarrow 0$

0 when $\epsilon \gg 1$

Differentially Private Mean Estimation

Definition - The Gaussian Mechanism [Dwork et al 2006]

Consider $f: \mathcal{X} \rightarrow \mathbb{R}^d$ and $\Delta_2 := \sup_{X, X' \in \mathcal{X}} \|f(X) - f(X')\|_2$, ↗ sensitivity

Then $A: X \rightarrow f(X) + \mathcal{N}\left(0, \frac{2\Delta_2^2 \log(1.25/\delta)}{\epsilon^2} \mathbf{I}_d\right)$ is (ϵ, δ) -DP

Example: Sample Mean

$$\mathcal{X} = \mathbb{B}_d(0,1), \quad f(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i, \quad \Delta_2 = \frac{2\sqrt{d}}{n}$$

↪ $A(x_1, \dots, x_n) = \frac{1}{n} \sum_i x_i + \mathcal{N}\left(0, \frac{2d \log(1.25/\delta)}{n} \mathbf{I}_d\right)$

↪ $\mathbb{E} \|A(x_1, \dots, x_n) - \frac{1}{n} \sum_i x_i\|^2 = \tilde{\Theta}\left(\frac{d^2}{n^2 \epsilon^2}\right)$

Example: Gaussian Mean Estimation (informal).

$$\text{Let } \Sigma \leq \mathbf{I}_d, \quad x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} N(0, \Sigma_d), \quad f(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i, \quad \Delta_2 \approx \sqrt{\text{tr} \Sigma} \log(n)$$

$$A(x_1, \dots, x_n) \approx \frac{1}{n} \sum_i \text{clip}(x_i; \Delta_2) + \mathcal{N}\left(0, \frac{\sqrt{d \text{tr} \Sigma} \log(n/\delta)}{n} \mathbf{I}_d\right)$$

statistical rate
adaptive to the effective
dimension

→ with probability ≥ 0.99 , $\|A(x_1, \dots, x_n)\|^2 \lesssim \frac{\text{tr}(\Sigma)}{n} + \frac{d \text{tr} \Sigma \log(1/\delta)}{n^2 \epsilon^2} + \log(n/\beta)$

↘ still dimension-
dependent!

A Curse of dimensionality in DP gaussian mean estimation

Setting:

- ▶ Take a prior distribution $\mathcal{D}$ on (μ, Σ)
- ▶ Compare
 - ▶ Error of « covariance-aware » (ϵ, δ) - DP algorithms
 - ▶ Error of « covariance-agnostic » (ϵ, δ) - DP algorithms

$$\mathfrak{R}_{\text{aware}} = \inf \left\{ \alpha^2 : \exists A \text{ } (\epsilon, \delta)\text{-DP, } \mathbb{P}_{\mu, \Sigma \sim \mathcal{D}} [\|A(X, \Sigma) - \mu\| \leq \alpha] \geq 2/3 \right\}$$

$X \sim \mathcal{N}(\mu, \Sigma)^{\otimes n}$

$$\mathfrak{R}_{\text{agnostic}} = \inf \left\{ \alpha^2 : \exists A \text{ } (\epsilon, \delta)\text{-DP, } \mathbb{P}_{\mu, \Sigma \sim \mathcal{D}} [\|A(X) - \mu\| \leq \alpha] \geq 2/3 \right\}$$

$X \sim \mathcal{N}(\mu, \Sigma)^{\otimes n}$

Theorem - Random Projection Σ

Prior: Σ uniformly random among the k -dimensional orthogonal projection, and $\mu \sim \mathcal{N}(0, c\mathbf{I}_d)$

Parameter: If $n < k$, $\delta \lesssim 1/d$ and $\epsilon = 1$, then:

low-sample regime: $n < \text{tr } \Sigma$

$$\frac{d}{n} + \frac{dk}{n^2} \approx \mathfrak{R}_{\text{agnostic}} \geq \frac{d}{k} \mathfrak{R}_{\text{aware}} \approx \frac{d}{k} \left(\frac{k}{n} + \frac{k^2}{n^2} \right)$$

Naive
algorithm

Dagan et al. 2024

Theorem - Heavy-Tailed Diagonal Σ

Prior: $\Sigma = \text{Diag}(\sigma_1, \dots, \sigma_d)$, with $1/\sigma_i \stackrel{\text{i.i.d.}}{\sim} \chi^2(1)$ and $\mu \sim \mathcal{N}(0, c\mathbf{I}_d)$

Parameter: $\log(d) \lesssim n \lesssim d^{1/8.5}$ and $\delta \lesssim 1/n$

high-sample regime:
 $n \gg \text{tr}(\Sigma)/\|\Sigma\|_2 \approx \log(d)$

$$\tilde{\Theta}\left(\frac{d\|\Sigma\|_2^2}{n^4}\right) \geq \mathfrak{R}_{\text{agnostic}} \geq \frac{\sqrt{d}}{n^{8.5}} \mathfrak{R}_{\text{aware}} \approx \frac{\sqrt{d}}{n^{8.5}} \frac{\|\Sigma\|_2^2}{n}$$

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unknown cov. algorithm

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$\gamma(\mathbf{L})$ measures the graph's connectivity

